

Zero- Versus Infinite-Temperature Damping in Variational Quantum Circuits: Feature Scale, Sampling Cost, and Frame Gauge

Vu-Quoc-Minh Nguyen ^{1,2} Tuan-Vu Truong ^{3,2} Hoang-Long Nguyen ² and Trung-Khanh Le ^{1,*}

¹*Faculty of Electronics and Telecommunications, University of Science, Vietnam National University Ho Chi Minh City (VNU-HCM), Ho Chi Minh City 70000, Vietnam*

²*Identity Quantum Computing JSC, 117 Tran Duy Hung Street, Hanoi 10000, Vietnam*

³*College of Natural Sciences, Can Tho University, 3-2 Road, Can Tho City 94000, Vietnam*

(Dated: September 30, 2026)

The Pauli twirl of amplitude damping (AD) is generalized amplitude damping at infinite temperature: it keeps the contraction of AD and removes its non-unital term, so comparing the two in variational circuits isolates the zero-temperature bias, which acts mainly through the scale of the features. For random parameters, features under AD settle on a floor, which at strong damping is set by the last layer, has a closed form, and at fixed T_1 falls with temperature as $\tanh(\hbar\omega/2k_B T)$; under the twirls they shrink by a constant factor per layer, up to eight qubits. A trainable output scale removes most of the resulting accuracy differences, leaving AD ahead of its twirls by at most about three percentage points in our simulations; what it removes reappears as a cost in measurement shots: trained and tested with 10^3 shots per image, a four-qubit classifier under AD at $p = 0.3$ stays within 1.5 points of noiseless accuracy, while the twirled classifiers lose up to 33. At weaker damping the separation depth grows roughly as $(np)^{-1} \ln(1/p)$. The damping direction is a gauge when the damping follows complete entangling layers and the circuit boundaries are trainable; in an eigensolver it becomes physical inside a decomposed two-qubit gate.

I. INTRODUCTION

Variational quantum algorithms and quantum machine learning models are built from parameterized quantum circuits and are expected to run on noisy intermediate-scale quantum devices [1–3]. Whether such models can be trained depends on how the output concentrates. Gradients of deep random circuits, and of global cost functions even in shallow ones, vanish exponentially with the number of qubits [4–6]. Unital noise adds a concentration with depth: the state approaches the maximally mixed state, and local expectation values and their gradients vanish exponentially [7, 8]. Non-unital noise behaves differently. Amplitude damping (AD), which models energy relaxation (T_1 decay) [9], a dominant source of noise in superconducting qubits [10], drives the state toward a pure state. Under non-unital noise, cost functions of local observables do not exhibit barren plateaus [11], and the output distribution of random circuits never anti-concentrates [12]. For general noise, the gradients with respect to early layers still vanish [13], and amplitude damping, a Hilbert–Schmidt-contractive map, need not produce noise-induced barren plateaus, although its cost concentrates onto a noise-induced limit set beyond logarithmic depth [14]. On hardware dominated by T_1 decay, gradient magnitudes saturate instead of decaying exponentially [15], and engineered dissipation can restore trainability [16].

Noise tailoring removes this structure. Randomized compiling, or Pauli twirling, turns an arbitrary gate error into a stochastic Pauli channel [17, 18], which is easier to characterize and mitigate [19, 20], and a Clifford

twirl turns it into depolarizing noise [21]. The Pauli twirl of AD keeps its contraction and discards its non-unital term: it is the equal mixture of damping toward $|0\rangle$ and toward $|1\rangle$, that is, generalized amplitude damping at infinite temperature. Comparing AD with its Pauli twirl in a variational model therefore asks what the zero-temperature bias of the damping does. Van Rossum *et al.* [22] made this comparison in data re-uploading circuits and in a variational eigensolver. They found that the twirled channels reduce the range of outputs the model can fit, which they use as a measure of expressivity, reduce gradient magnitudes, and degrade eigensolver solutions, and that in the eigensolver damping in the reversed direction performs significantly worse than ordinary damping. They concluded that “commonly used noise reshaping techniques, such as Pauli twirling, may inadvertently degrade the performance of variational algorithms” [22].

Here, we show that the bias acts on variational models mainly through the scale of their features, and that its direction is a gauge when the damping acts after complete entangling layers and trainable gates and the readout absorb the frame at the boundaries of the circuit. For random parameters, local features under AD settle on a floor, which at strong damping is set by the last layer and has a closed form, whereas under the twirls they shrink by a constant factor per layer; we establish this at widths up to eight qubits. A trainable affine readout absorbs a contraction, so it removes most of the differences in fit and accuracy in exact simulation [23, 24]. What it removes reappears as a cost in measurement shots, which we measure: trained and tested with 10^3 shots per image, the classifier under AD stays close to the noiseless accuracy, whereas the twirled classifiers lose up to 33 percentage points. The separation needs a depth that

* Contact: ltkhanh@hcmus.edu.vn

grows roughly as $(np)^{-1} \ln(1/p)$ for n qubits at damping strength p , so at the damping of current hardware it is a property of very deep circuits. At finite temperature and fixed T_1 the floor falls as $\tanh(\hbar\omega/2k_B T)$, and the Pauli twirl is its infinite-temperature end. The direction of the damping, in turn, is a label that a change of frame can move. We give conditions under which reversing it is a reparametrization of the model, implemented by a Pauli frame or by spin time reversal. The frame propagation is that of the parameter symmetries of Ref. [25]; here it relates two noise models. Time reversal leaves every single-qubit rotation invariant, so the encoding never protects the direction. Only the fixed multi-qubit gates, the placement of the noise relative to them, and the boundaries of the circuit can make the direction physical. Where the frame is absorbed, the initialization is symmetric, and the optimizer is equivariant, trained models under the two directions are identically distributed.

We test these statements on three models: the single-qubit re-uploading fits of Ref. [22], a four-qubit re-uploading classifier on MNIST and Fashion-MNIST, and the three-qubit eigensolver of Ref. [22]. With a trainable output scale, AD and its twirls fit equally well and classify to within about one percentage point at four layers of strong damping. The direction effect in the eigensolver exists because the damping acts inside the decomposed two-qubit gate, it vanishes exactly when the same damping follows the gate, and it persists when the transverse field dominates; on an open chain the sign of the coupling is itself a gauge of the ansatz, so the effect cannot depend on it. Beyond the scale, the bias leaves AD ahead of its twirl by up to three percentage points in exact simulation; at $p = 0.1$ this margin grows from 0.9 points at four layers to 2.6 at the separation depth. At three and four layers it also gives the classifier a small gain over the noiseless circuit in exact simulation, which does not survive training and testing with 10^3 shots per image, and on MNIST it keeps the input sensitivity from growing with depth; on Fashion-MNIST it does not.

The paper is organized as follows. Section II introduces the channels and the three models. Section III treats the output scale and Sec. IV the gauge. Section V shows how the non-unital term sets the feature scale and the sampling cost, and what it changes beyond them, and Sec. VI concludes.

II. CHANNELS AND MODELS

A. Channels

A single-qubit channel with Kraus operators K_k ,

$$\mathcal{N}(\rho) = \sum_k K_k \rho K_k^\dagger, \quad (1)$$

acts on the Bloch vector as an affine map [26],

$$\mathbf{r} \mapsto T\mathbf{r} + \mathbf{t}, \quad (2)$$

$$T_{ij} = \frac{1}{2} \text{Tr}[P_i \mathcal{N}(P_j)], \quad t_i = \frac{1}{2} \text{Tr}[P_i \mathcal{N}(I)], \quad (3)$$

where $P_i \in \{X, Y, Z\}$. The channel is unital, $\mathcal{N}(I) = I$, if and only if $\mathbf{t} = \mathbf{0}$. We compare the conditions of Table I, as in Ref. [22]. Amplitude damping of strength p has

$$T_{\text{AD}} = \text{diag}(c, c, c^2), \quad \mathbf{t}_{\text{AD}} = (0, 0, p), \quad (4)$$

with $c = \sqrt{1-p}$. Its Pauli twirl [17, 27] has the same contraction and no non-unital term,

$$T_{\text{Pauli}} = T_{\text{AD}}, \quad \mathbf{t}_{\text{Pauli}} = \mathbf{0}, \quad (5)$$

and its Clifford twirl is the depolarizing channel [21],

$$T_{\text{Depol}} = \lambda_d \mathbb{I}_3, \quad \mathbf{t}_{\text{Depol}} = \mathbf{0}, \quad (6)$$

with $\lambda_d = \frac{1}{3}(2c + c^2)$, the mean of the contraction factors of AD. The reversed channel damps toward $|1\rangle$,

$$\text{AD}^{\text{flip}} = \mathcal{X} \circ \text{AD} \circ \mathcal{X}, \quad \mathcal{X}(\rho) = X\rho X, \quad (7)$$

with the same T and $\mathbf{t} = (0, 0, -p)$. Since $\text{ZADZ} = \text{AD}$ and $\text{XADX} = \text{YADY} = \text{AD}^{\text{flip}}$, the Pauli twirl is the equal mixture

$$\text{AD}_{\text{Pauli}} = \frac{1}{2}(\text{AD} + \text{AD}^{\text{flip}}), \quad (8)$$

which is generalized amplitude damping at infinite temperature [26]. Comparing AD with its Pauli twirl therefore compares damping at zero and at infinite temperature. More generally, generalized amplitude damping with excited-state population N has the same T and $\mathbf{t} = (0, 0, p(1 - 2N))$; the frame $\mathcal{X}$ maps N to $1 - N$, and AD, its Pauli twirl, and AD^{flip} are the cases $N = 0, \frac{1}{2}$, and 1. As in Ref. [22], the twirled conditions replace the channel by its twirl at every insertion; no twirling protocol implements this after non-Clifford gates, so the twirled channels serve as model channels with the contraction of AD. In simulations of error correction, this Pauli twirling approximation is accurate for some stabilizer circuits [28] but misses features of amplitude damping that other efficiently simulable channels capture [29].

B. Single-qubit re-uploading fits

Our first model reproduces the setting of Fig. 2(a) of Ref. [22]. A single qubit starts in $|0\rangle$ and passes through $L = 2$ layers. With $\mathcal{W}_l = \mathcal{U}(R_Z R_Y R_Z)$ the trainable blocks, $\mathcal{S}_x = \mathcal{U}(R_X(x))$ the encoding, and $\mathcal{U}(V): \rho \mapsto V\rho V^\dagger$, the channel acts after every block,

$$\Phi_{x,\theta} = \mathcal{N} \circ \mathcal{W}_2 \circ \mathcal{N} \circ \mathcal{S}_x \circ \mathcal{N} \circ \mathcal{W}_1 \circ \mathcal{N} \circ \mathcal{S}_x \circ \mathcal{N} \circ \mathcal{W}_0, \quad (9)$$

with five insertions and nine parameters. The output of the raw model is

$$f(x) = \text{Tr}[Z \Phi_{x,\theta}(|0\rangle\langle 0|)]. \quad (10)$$

TABLE I. Noise conditions at strength p , with $c = \sqrt{1-p}$ and $\Lambda = 1 - \frac{1}{3}(2c + c^2)$. Unital channels are given by their Pauli weights, with $p_0 = 1 - \sum_{k \neq 0} p_k$.

channel	Kraus operators	$(T, \mathbf{t})$
None	$K_0 = I$	$\mathbb{I}_3, \mathbf{0}$
AD	$\text{diag}(1, c), \sqrt{p} 0\rangle\langle 1 $	$\text{diag}(c, c, c^2), (0, 0, p)$
AD ^{flip}	$\text{diag}(c, 1), \sqrt{p} 1\rangle\langle 0 $	$\text{diag}(c, c, c^2), (0, 0, -p)$
Pauli	$p_x = p_y = \frac{p}{4}, p_z = \frac{1}{4}(2 - p - 2c)$	$\text{diag}(c, c, c^2), \mathbf{0}$
Depol	$p_x = p_y = p_z = \frac{\Lambda}{4}$	$\frac{1}{3}(2c + c^2) \mathbb{I}_3, \mathbf{0}$

As targets we use degree-2 trigonometric polynomials,

$$g'(x) = a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x, \quad (11)$$

with a_k, b_k drawn uniformly from $[-1, 1]$, normalized to unit range and zero mid-range and then scaled to a range R , and sampled at 250 points in $[-2\pi, 2\pi]$. Ref. [22] measures expressivity as the largest R for which Adam (learning rate 0.20, $\beta_1 = 0.45$) reaches a mean squared error below 10^{-5} within 150 steps. We use 200 seeded targets, the same for every channel, and compare the raw model with a rescaled model,

$$f_{s,b}(x) = s f(x) + b, \quad (12)$$

in which s and b are trained together with the circuit. The details of our reading of the protocol are given in Appendix A.

C. Four-qubit re-uploading classifier

Our second model is a data re-uploading classifier [30, 31] on $n = 4$ qubits (Fig. 1). We standardize each MNIST image [32] of the digits $\{1, 3, 5\}$, reduce it to four principal components u_j , and map each component to an angle by a min-max transformation,

$$a_j = 2\pi \frac{u_j - u_j^{\min}}{u_j^{\max} - u_j^{\min}}. \quad (13)$$

The standardization, the principal components, and the extrema are all fitted on 1000 training images and applied unchanged to all 3037 test images of the three digits; 14 of the 12 148 test angles fall outside $[0, 2\pi]$ and are clipped. With the trainable blocks, the encoding, and the ring entangler,

$$\mathcal{R}^{(\ell)} = \bigotimes_{j=1}^n R_Z R_Y R_Z, \quad S(\mathbf{a}) = \bigotimes_{j=1}^n R_X(a_j), \quad (14)$$

$$E = \prod_{j=1}^n \text{CNOT}_{j,j+1}, \quad (15)$$

with the qubit index taken modulo n , a layer performs

$$\mathcal{L}_{\mathbf{a}}^{(\ell)} = \mathcal{N}^{\otimes n} \circ \mathcal{U}(S(\mathbf{a})) \circ \mathcal{N}^{\otimes n} \circ \mathcal{U}(E \mathcal{R}^{(\ell)}), \quad (16)$$

and the circuit is

$$\Phi_{\mathbf{a},\theta} = \mathcal{N}^{\otimes n} \circ \mathcal{U}(\mathcal{R}^{(L)}) \circ \mathcal{L}_{\mathbf{a}}^{(L-1)} \circ \dots \circ \mathcal{L}_{\mathbf{a}}^{(0)}, \quad (17)$$

with $2L + 1$ insertions of the channel per qubit. The features are the single-qubit expectation values

$$z_j(\mathbf{a}, \theta) = \text{Tr}[Z_j \Phi_{\mathbf{a},\theta}(|0\rangle\langle 0|^{\otimes n})], \quad (18)$$

and the scores of the three classes are

$$\ell(\mathbf{a}) = \begin{cases} W\mathbf{z} + \mathbf{b} & \text{(raw),} \\ W \text{diag}(\boldsymbol{\sigma})^{-1}(\mathbf{z} - \boldsymbol{\mu}) + \mathbf{b} & \text{(standardized),} \end{cases} \quad (19)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are the mean and the population standard deviation of the features over the $M = 1000$ training images,

$$\mu_j = \frac{1}{M} \sum_{m=1}^M z_j(\mathbf{a}_m, \theta), \quad (20)$$

$$\sigma_j^2 = \frac{1}{M} \sum_{m=1}^M [z_j(\mathbf{a}_m, \theta) - \mu_j]^2. \quad (21)$$

They are recomputed at every step, with gradients flowing through them, and the values at the final parameters are applied to the test images. The standardized readout has no parameter beyond those of the raw one; it only fixes the scale at which W acts. No regularizing constant is added to $\boldsymbol{\sigma}$; in one run a feature's spread fell to 5×10^{-5} and the training diverged (Table II). Training minimizes the softmax cross-entropy with Adam [33] at learning rate 0.05 for 100 full-batch steps unless stated otherwise, without weight decay. The circuit angles start uniformly in $[0, 2\pi]$, the readout weights from $0.5\mathcal{N}(0, 1)$, and the bias at zero. We use $p = 0.3$ and simulate the density matrix exactly in PyTorch [34], so all expectation values are free of sampling error. As a second data set we use the Fashion-MNIST [35] classes T-shirt, pullover, and coat, with 3000 test images. Three classes and 1000 training images keep exact density-matrix training of eight initializations per condition affordable. For reference, logistic regression, a k -nearest-neighbor classifier ($k = 15$), and a support-vector machine with a radial kernel, trained on the same angles, reach 0.856, 0.879, and 0.886 on MNIST and 0.733, 0.758, and 0.748 on Fashion-MNIST.

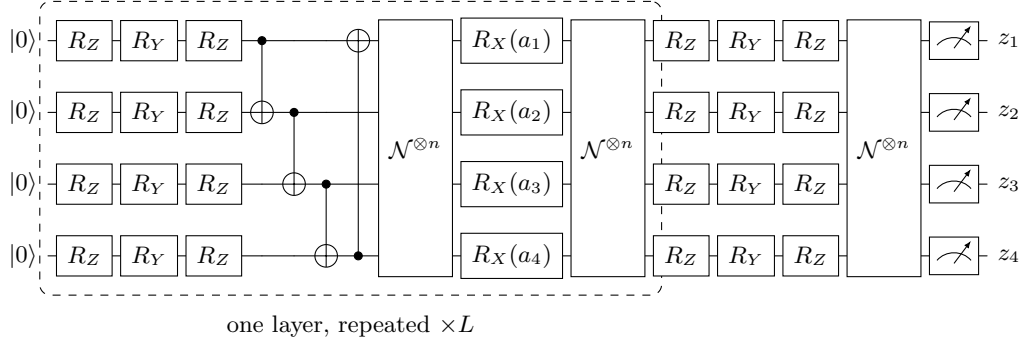


FIG. 1. The classifier at $n = 4$, drawn for one layer and the final block. Each $R_Z R_Y R_Z$ triple carries three trainable angles; the encoding column re-uploads the angle vector $\mathbf{a}$ in each of the L layers. The channel acts on every qubit after the entangler, after the encoding, and after the final block, giving $2L + 1$ insertions per qubit. The features $z_j = \langle Z_j \rangle$ enter a trainable linear readout.

We quantify the input sensitivity of a trained classifier by

$$\chi = \langle \|\nabla_{\mathbf{a}} \ln q_y(\mathbf{a})\| \rangle, \quad (22)$$

where q_y is the softmax probability of the true class y and the average runs over the test images.

For the finite-shot runs, each image is measured N_s times in the computational basis, and $\hat{z}_j$ is the mean of the N_s outcomes ± 1 of qubit j . At test time we sample the outcomes from the exact distribution of the measured state, five times per model, and in every draw we re-estimate $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ from N_s shots of each training image, as one would on hardware. In training we replace each feature by

$$\hat{z}_j = z_j + \sqrt{\frac{1 - z_j^2}{N_s}} \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, 1), \quad (23)$$

redrawn at every step and for every image, and compute $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ from the noisy features; gradients flow through z_j . This Gaussian model has the exact variance of a single feature but neglects the correlations between qubits, and its gradients are those of the noisy loss, without the additional shot noise of parameter-shift estimates. We train with $N_s = 10^3$ at $L = 1$ to 4, and with $N_s = 10^2$ and 10^4 at $L = 4$. We define the operational shot cost of a trained model as the smallest N_s at which its mean test accuracy lies within 0.01 of its accuracy with exact expectation values, interpolated in $\log N_s$ on a grid from 10 to 10^8 , and we report its geometric mean over the initializations. It is undefined for three exactly trained models at $L = 1$, which we omit: the diverged Pauli run, and one noiseless and one depolarizing model whose accuracy stays more than 0.01 below its exact value up to 10^8 shots.

D. Three-qubit variational eigensolver

Our third model is the eigensolver of Ref. [22] for the periodic transverse-field Ising chain on three qubits,

$$H = -J \sum_{i=1}^3 Z_i Z_{i+1} - h \sum_{i=1}^3 X_i, \quad (24)$$

with $J = 1$, $h = 0.5$, and $Z_4 \equiv Z_1$, whose ground-state energy is $E_0 = -3.2321$. A trainable R_Y on each qubit is followed by four blocks. Each block applies $R_{ZZ}(\phi)$ to the three pairs and a trainable R_X to each qubit, and each two-qubit gate is decomposed as

$$R_{ZZ}(\phi) = \text{CNOT}_{i,i+1} [I \otimes R_Z(\phi)] \text{CNOT}_{i,i+1}. \quad (25)$$

We compare two noise placements with the same number of insertions. In the *inside* placement of Ref. [22], the channel acts on both qubits after each CNOT. In the *after* placement, it acts twice on both qubits after the complete R_{ZZ} . Single-qubit gates are noiseless. We minimize the energy and report the relative error,

$$E(\theta) = \text{Tr}[H\rho(\theta)], \quad \epsilon = \frac{E(\theta) - E_0}{|E_0|}, \quad (26)$$

with Adam (learning rate 0.05, 500 steps) from 50 initializations per point, drawn uniformly from $[0, 2\pi)$, with every channel of a point starting from the same parameters. Here and below we denote the noise strength of the eigensolver by γ , the p of Table I. For the inside placement, our implementation reproduces the energies of the Qiskit [36] code that the authors of Ref. [22] deposited openly [37] to 4×10^{-14} for the noiseless circuit, AD, reversed AD, and both twirls, at $\gamma \in \{0.01, 0.05, 0.1, 0.2\}$.

E. Statistics

Classifier results are given as the mean and standard deviation (SD) over eight initializations with fixed data.

All runs share one test set, so a comparison between two conditions is also uncertain because the test set is finite. For the comparisons on which our conclusions rest, we therefore quote a 95% confidence interval (CI) from a two-level bootstrap [38] that resamples both the initializations, jointly for the two conditions since they share initial parameters, and the test images. The hypothesis tests behind every comparison are collected in Appendix C.

III. OUTPUT SCALE

A channel acts on a variational model first through the scale of its output. For depolarizing noise on one qubit this is its only effect. Between AD and its Pauli twirl, which share the contraction matrix T , the difference in scale is itself an effect of the non-unital vector $\mathbf{t}$ (Sec. V). Here we show what a trainable output scale removes and what it costs.

A. Scale and sampling cost

Suppose first that each noisy feature is an affine function of the noiseless one,

$$\tilde{z}_j = s z_j + b_j. \quad (27)$$

Then a trainable affine readout realizes on $\tilde{\mathbf{z}}$ every classifier it realizes on $\mathbf{z}$, and the noise has no effect on the model class. For a single qubit under depolarizing noise this holds exactly, since the channel commutes with every unitary: with N insertions,

$$\tilde{f} = \lambda_d^N f. \quad (28)$$

For anisotropic or non-unital channels, and on several qubits, the noisy model class is not a rescaled copy of the noiseless one, and whether the scale accounts for the differences is an empirical question.

The rescaling is not free on hardware, as we now show. A feature estimated from N_s shots, $\hat{z}_j$, has the variance

$$\text{Var}(\hat{z}_j) = \frac{1 - z_j^2}{N_s} \leq \frac{1}{N_s}. \quad (29)$$

The number of shots is a leading cost of variational algorithms [39, 40]. The readout divides the feature by its scale, so it amplifies the shot noise by the same factor. Keeping the class scores at a fixed precision therefore requires a number of shots proportional to the inverse square of the feature scale. We report the shot factor

$$\kappa(\mathcal{N}) = \left(\frac{\sigma(\text{None})}{\sigma(\mathcal{N})} \right)^2, \quad (30)$$

where σ is the mean over features and over the initializations of the standard deviation of z_j over the training

images at the trained parameters. For the single-qubit fits we use $1/|s|$, with s the median fitted output scale, in place of σ , so that $\kappa = (s/s_{\text{None}})^2$. In the language of error mitigation, the trained readout performs a rescaling [24, 41], which cannot remove an exponential concentration without an exponential number of shots [23]. In Sec. V A we compare κ with the operational shot cost defined in Sec. II C.

B. Single-qubit fits

With the raw output and a free target offset, we recover the ordering of Ref. [22] at $p \geq 0.1$ [Fig. 2(a)]: at $p = 0.1$ the largest converging range is 0.864 for AD, against 0.769 and 0.745 for the Pauli and Clifford twirls. The metric is not monotone in the noise, however. The noiseless model reaches only 0.646, below every noisy channel at $p \leq 0.1$, so the metric measures convergence within the 150-step budget as much as the range the model can reach; with targets centered on zero, the ordering at $p = 0.1$ reverses (Appendix C).

With a trained scale and offset, we find that the difference disappears [Fig. 2(b)]. At $R = 1$, AD and the Pauli twirl converge on 96.5 to 99.0% and on 98.5 to 99.0% of the targets within 1000 steps at $p = 0.05$ to 0.2, the noiseless model on 98.5%, and none of the paired differences is significant; within 150 steps the only significant difference favors the Pauli twirl (Appendix C). What the twirl changes is the scale [Fig. 2(c)]. At $p = 0.2$ the Pauli twirl needs 1.24 to 1.57 times the shots of AD at the two ranges, and the Clifford twirl, whose shot factor follows the exact depolarizing value λ_d^{-10} , 0.92 to 1.16 times. On one qubit the cost of twirling is therefore modest; it becomes large in the classifier, where it compounds over qubits and layers. The same scale accounts for the smaller gradient magnitudes reported in Ref. [22], which notes that under Pauli noise the loss gradients carry the attenuation factors of the output. For a rescaled output, $\partial(sf)/\partial\theta = s \partial f/\partial\theta$, so a trained s restores the gradient magnitude, but not the signal-to-noise ratio of shot-estimated gradients, whose noise it amplifies by the same factor; estimating the gradients to fixed precision costs the shot factor κ .

C. Classifier

With the raw readout and 100 training steps, we find that the unital channels appear to lose trainability with depth [Table II, Fig. 3(a)]. Between $L = 1$ and $L = 4$ the test accuracy falls from 0.847 to 0.715 for the Pauli twirl and from 0.851 to 0.695 for depolarizing noise, and the training accuracy falls with it (to 0.700 and 0.672), while AD stays at 0.84 and the noiseless circuit at 0.86. A reading in terms of trainability would conclude that the non-unital term keeps the deep circuit trainable.

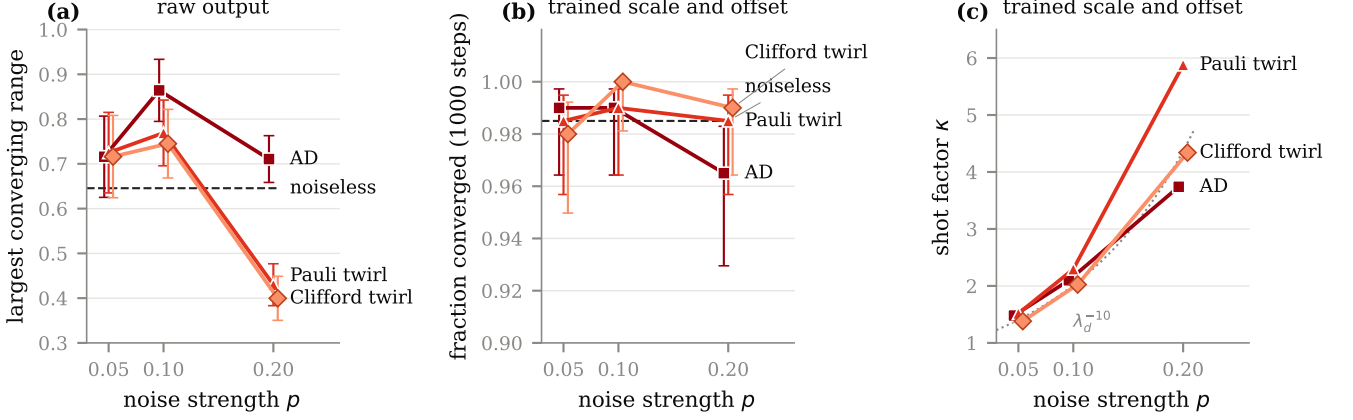


FIG. 2. Single-qubit re-uploading fits of Ref. [22] (200 targets). (a) Largest converging output range of the raw model $f = \langle Z \rangle$, with a free target offset as in Ref. [22]. (b) Fraction of targets of range $R = 1$ that converge within 1000 steps when an output scale s and offset b are trained. (c) Shot factor $\kappa = (s/s_{\text{None}})^2$ from the median fitted scale; the dotted line is the exact depolarizing value λ_d^{-10} . Bars are 95% confidence intervals of the mean over targets in (a) and Wilson 95% intervals in (b).

TABLE II. Test accuracy of the classifier on 3037 MNIST images (mean $\pm$ SD over eight initializations) against the number of re-uploading layers L . The raw readout is trained for 100 steps and, at $L = 4$, for 1000 steps; the standardized readout of Eq. (19) for 100 steps. [†]One of the eight runs diverged (test accuracy 0.382); the other seven lie between 0.860 and 0.865.

	raw readout					standardized readout				
	$L = 1$	$L = 2$	$L = 3$	$L = 4$	$L = 4, 1000 \text{ steps}$	$L = 1$	$L = 2$	$L = 3$	$L = 4$	
None	.861 $\pm$.003	.866 $\pm$.005	.865 $\pm$.009	.863 $\pm$.004	.862 $\pm$.007	.860 $\pm$.005	.869 $\pm$.007	.868 $\pm$.006	.869 $\pm$.005	
AD	.842 $\pm$.010	.835 $\pm$.032	.839 $\pm$.005	.841 $\pm$.004	.872 $\pm$.003	.861 $\pm$.003	.872 $\pm$.003	.877 $\pm$.004	.876 $\pm$.005	
Pauli	.847 $\pm$.003	.817 $\pm$.039	.787 $\pm$.044	.715 $\pm$.057	.837 $\pm$.013	.802 $\pm$.170 [†]	.873 $\pm$.005	.869 $\pm$.009	.870 $\pm$.007	
Depol	.851 $\pm$.003	.814 $\pm$.041	.778 $\pm$.034	.695 $\pm$.041	.826 $\pm$.012	.860 $\pm$.005	.869 $\pm$.006	.869 $\pm$.010	.870 $\pm$.004	

We find that the readout explains it. Trained for 1000 steps with the same raw readout, the unital channels recover to 0.837 and 0.826 at $L = 4$, while the norm of their readout weights grows from about 19 at step 100 to 106 and 109 at step 1000; for the noiseless circuit it grows from 8 to 20. With the standardized readout, every channel reaches the accuracy of the noiseless circuit at every depth [Fig. 3(b)]. The Pauli and depolarizing channels are then indistinguishable from it: at $L = 4$ the differences are +0.001 (95% CIs $[-0.007, +0.009]$ and $[-0.006, +0.008]$), and no difference at $L = 2$ or 3 exceeds 0.004. At 100 steps the raw readout cannot grow its weights fast enough to use features whose scale the unital channels have shrunk to 4 and 5 percent of the noiseless scale after 100 steps (and to below 1 percent before training); the circuit itself remains trainable. At $L = 1$ the depolarizing channel provides an exact check. Each feature then crosses all three noise insertions as a weight-one Pauli string, so the channel multiplies every feature by the same factor for every θ ,

$$z_j^{\text{Depol}}(\mathbf{a}, \theta) = \lambda_d^3 z_j^{\text{None}}(\mathbf{a}, \theta). \quad (31)$$

The standardized readout removes a uniform scale, and the depolarizing and noiseless classifiers indeed make

identical predictions on every test image. AD is not a rescaling, and it ends slightly above the noiseless circuit at $L = 3$ and 4 (+0.009, CI $[+0.003, +0.015]$, and +0.008, CI $[+0.002, +0.013]$), but not at $L \leq 2$ (Sec. V). The same holds on Fashion-MNIST at $L = 4$. With the raw readout, the unital channels are 16 and 17 points below the noiseless circuit (0.765) and AD is 5 points below it. With the standardized readout, the unital channels are not resolved from the noiseless circuit, and AD is above it by 0.017, CI $[+0.007, +0.029]$ (Appendix C). The shot factors are 34 for AD and 9.9×10^3 and 8.2×10^3 for the Pauli and depolarizing channels.

What survives, as we show in Fig. 3(d), is the feature scale. At the end of training with the standardized readout, the shot factor grows by a factor of about 14 per layer for the unital channels, from 4.1 at $L = 1$ to 1.1×10^4 at $L = 4$ for depolarizing noise, and to 1.6×10^4 for the Pauli twirl. For AD it grows by a factor of about 2.4 per layer, from 3.6 to 50. The shot factor depends on the training objective and is not a property of the channel alone. With the raw readout the optimizer moves the parameters toward less contracted features: after 1000 steps at $L = 4$ the factors are 11 for AD, 320 for the Pauli twirl, and 470 for depolarizing noise. The mea-

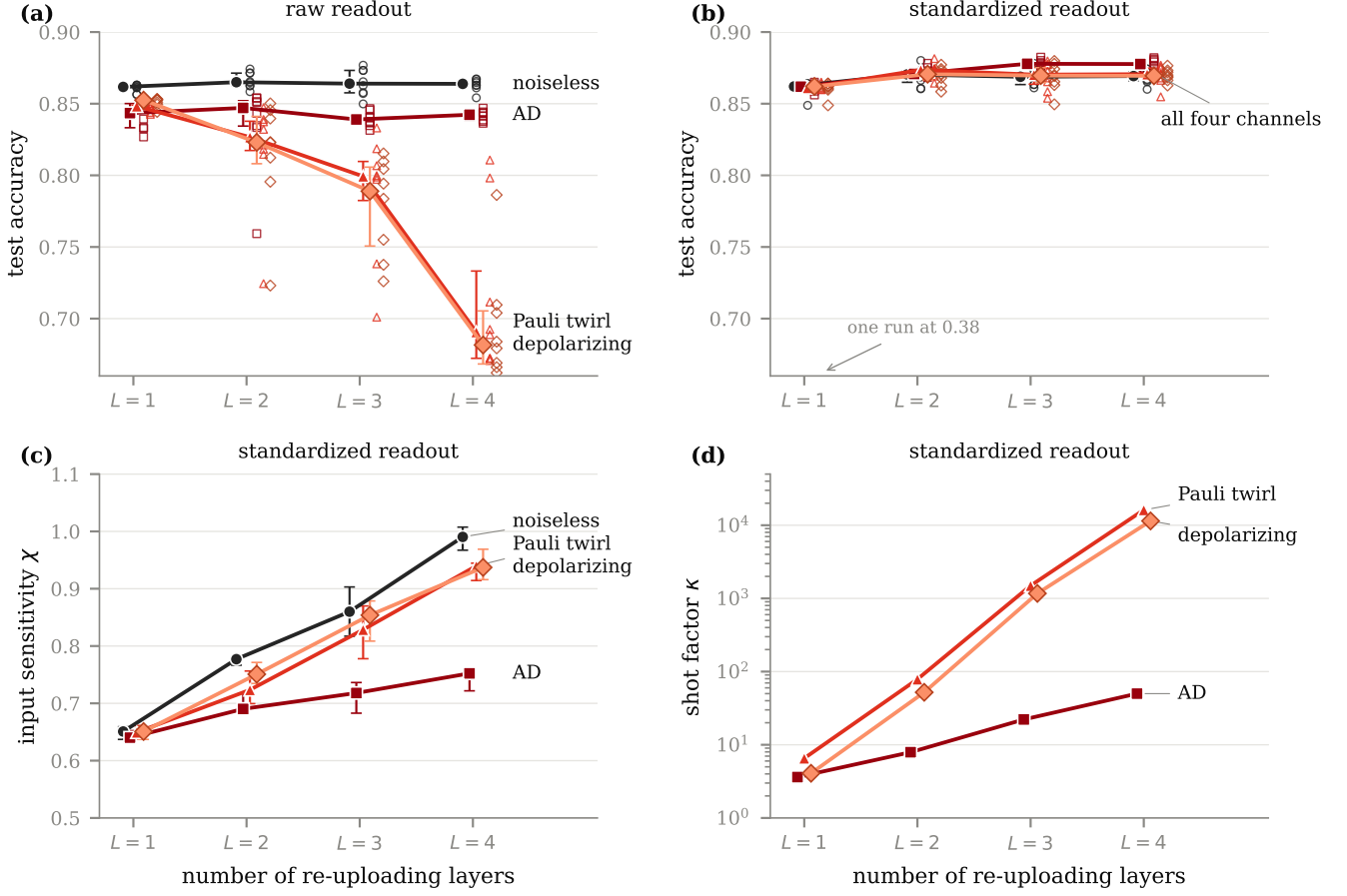


FIG. 3. Four-qubit classifier on MNIST against the number of re-uploading layers. (a) Test accuracy with the raw readout and (b) with the standardized readout, both after 100 steps. (c) Input sensitivity and (d) shot factor κ of Eq. (30) with the standardized readout. Panels (a)–(c) show the median and interquartile range over eight initializations, with each run as a hollow marker; one Pauli run at $L = 1$ in (b) diverged and lies below the axis.

sured shot costs follow κ for the unital channels but not for AD (Sec. V A).

IV. GAUGE FREEDOM OF THE DAMPING DIRECTION

A. When the damping direction is a gauge

Amplitude damping toward $|1\rangle$ instead of $|0\rangle$, AD^{flip} of Eq. (7), has the same contraction matrix $T = \text{diag}(c, c, c^2)$ as AD and the opposite non-unital vector, $\mathbf{t} = (0, 0, -p)$. Whether a circuit can tell the two apart is not a property of the channel but of where the circuit lets a Pauli frame travel.

Write the reversed circuit as $\Phi^{\text{rev}} = E_m^{\text{rev}} \circ \dots \circ E_1^{\text{rev}}$, where the elements E_k^{rev} are its gates, encodings, and noise layers. We insert $g_k \circ g_k = \text{id}$ at every cut k between the elements. Each g_k is a Pauli frame [42]: a Pauli conjugation, or its composition with complex conjugation

in the computational basis,

$$g_k(\rho) = \begin{cases} P_k \rho P_k & (\text{unitary type}), \\ P_k \bar{\rho} P_k & (\text{antiunitary type}), \end{cases} \quad (32)$$

with all g_k of one type. The reversed circuit becomes

$$\Phi^{\text{rev}} = g_m \circ \tilde{E}_m \circ \dots \circ \tilde{E}_1 \circ g_0, \quad (33)$$

$$\tilde{E}_k = g_k \circ E_k^{\text{rev}} \circ g_{k-1}. \quad (34)$$

Proposition 1 (Frame gauge). *Reversing the damping is a reparametrization of the model,*

$$y^{\text{rev}}(x; \theta) = \eta y(x; \theta') \quad (35)$$

for all inputs x , with $\eta = \pm 1$, whenever a frame assignment of one type exists such that (i) at every noise location the frame acts as X or Y on each noisy qubit; (ii) every fixed gate maps the incoming frame to a Pauli frame; (iii) every data-encoding gate is left invariant; (iv) every trainable element is mapped back into its own family; (v) the input state is absorbed by the first element

on each qubit, and the final frame maps each measured observable to η times itself, with $\eta = -1$ allowed only if the readout has a learnable sign.

The proof is given in Appendix B.

Propagating Pauli operators through a parametrized circuit is the σ -pulse method of Fontana *et al.* [25], who used it to find parameter symmetries of a single model. They showed that these symmetries survive unital Pauli noise and can be broken by non-unital noise such as amplitude damping. Where a propagated frame crosses the channel as X or Y , it maps AD onto AD^{flip} , and this is how the symmetry breaks. Proposition 1 reads the same propagation as a map between two noise models, adds the antiunitary frames, and states the conditions on the encoding, the input, and the readout under which the map is a reparametrization. Noise can also lift degeneracies among the parameters of overparametrized circuits [43].

The antiunitary case contains spin time reversal,

$$\Theta = \bigotimes_q (-iY_q) K, \quad (36)$$

with K complex conjugation in the computational basis. It reverses every Pauli matrix and the imaginary unit,

$$\Theta \sigma_q \Theta^{-1} = -\sigma_q, \quad \Theta i \Theta^{-1} = -i, \quad (37)$$

so every single-qubit rotation is time-reversal invariant, while the damping is reversed because the Kraus operators of AD are real,

$$\Theta R_{\mathbf{n}}(a) \Theta^{-1} = R_{\mathbf{n}}(a), \quad \Theta \text{AD} \Theta^{-1} = \text{AD}^{\text{flip}}, \quad (38)$$

for any axis $\mathbf{n}$. Condition (iii) therefore never fails for single-qubit encodings, whatever their axis: only multi-qubit fixed gates, the placement of noise relative to them, and the two boundaries can break the gauge. In our classifier (noise after the whole entangling ring, a general trainable layer before it, an affine readout) all five conditions hold for R_X , R_Y , R_Z and a generic encoding axis alike, and the reversal is exact reparametrization to within 3.3×10^{-16} (Table III, at $n = 3$ and $L = 2$; with the explicit maps of App. B, to 9.4×10^{-16} for $n = 2$ to 4 and $L = 1$ to 4). A unitary Pauli frame suffices only when it commutes with the encoding generator and still flips the damping, which singles out X for R_X and Y for R_Y ; for R_Z or a generic axis only the antiunitary frame works. An encoding choice therefore never protects the damping direction from being a gauge. The damping-direction comparison of Sec. IV B therefore carries no information about the sign of the non-unital term.

Corollary 1 (Deterministic frames are gauge). *Under the conditions of Proposition 1, a deterministic relabelling of the noise by a Pauli frame or by time reversal leaves the reachable model class unchanged. If, in addition, $\theta \mapsto \theta'$ preserves the distribution of the initial parameters and the optimizer is equivariant under it, the trained models under AD and AD^{flip} are identically distributed.*

Both conditions hold for our classifier. For each frame of Table III, every Euler angle of the trainable layers maps as $\theta_k \mapsto \pm\theta_k + m_k\pi$ with $m_k \in \{0, 1\}$, and the readout as $W \mapsto -W$, $b \mapsto b$. These maps leave the initialization invariant, since the model depends on each angle only modulo 2π . The standardized readout maps $\mu \mapsto -\mu$ and $\sigma \mapsto \sigma$, and Adam without weight decay is equivariant under sign flips and translations of the coordinates. Trained classifiers under AD and AD^{flip} are therefore identically distributed over initializations; the same holds with shot noise, which is symmetric under $z \mapsto -z$. In the eigensolver of Ref. [22] with the channel moved after each complete R_{ZZ} (Sec. IV C), the frame changes only the initial R_Y angles, $y \mapsto \pi - y$, which preserves their initialization $U(0, \pi)$. The optimizer of Ref. [22], COBYLA, builds its first simplex along the positive coordinate axes and is not equivariant under this reflection, so there only the reachable energies are guaranteed to coincide. With the channel after each CNOT, as in Ref. [22], no frame exists (Table III).

Corollary 2 (The Pauli twirl is not frame-equivalent to AD). *Pauli twirling replaces AD by a convex mixture of frame-equivalent channels,*

$$\text{AD}_{\text{Pauli}} = \frac{1}{4} \sum_{P \in \{I, X, Y, Z\}} P \text{AD} P, \quad (39)$$

which is Eq. (8). It is not itself frame-equivalent to AD: every (anti)unitary Pauli frame preserves $\|\mathbf{t}\|$, which is p for AD and 0 for the twirl.

The single-qubit fits of Ref. [22] (Sec. III B; general single-qubit blocks, R_X encoding, noise after every block, output $f = \langle Z \rangle$ used directly) fail only condition (v): the frame left after the last channel flips the sign of $\langle Z \rangle$ and nothing absorbs it, so

$$f^{\text{rev}}(x; \theta) = -f(x; \theta'). \quad (40)$$

The two model classes are mirror images and differ as sets, since the last channel confines the output under AD and under reversed AD to

$$f(x; \theta) \in [2p - 1, 1], \quad f^{\text{rev}}(x; \theta) \in [-1, 1 - 2p]; \quad (41)$$

both upper bounds are attained. A target ensemble symmetric under $f \rightarrow -f$, or a trainable output scale, makes the two channels statistically indistinguishable.

B. Classifier

In the classifier of Fig. 1, time reversal maps each CNOT of the ring to

$$\Theta \text{CNOT} \Theta^{-1} = -\text{CNOT}(Z \otimes X), \quad (42)$$

which leaves a local Pauli residue on its input side. We can move this residue before the ring and absorb it into

TABLE III. Frame gauge for reversed amplitude damping, from an exhaustive search over all Pauli frames of each type (4^n strings; $n \leq 3$), with $p = 0.3$. “Gauge” means a frame assignment satisfying (i)–(v) exists; the explicit identity $y^{\text{rev}}(x; \theta) = \eta y(x; \theta')$ is then checked on random parameters and inputs, and the largest deviation is quoted. “None” means no frame of that type satisfies (i)–(v). The eigensolver rows place the channel once after each whole R_{ZZ} or after each CNOT; applying it twice after each R_{ZZ} , as in Fig. 4(a), leaves the frame assignment unchanged.

model	Pauli frame	time reversal / antiunitary
classifier, R_X encoding, noise after ring	gauge, 2.2×10^{-16}	gauge, 3.3×10^{-16}
classifier, R_Y encoding	gauge (Y frame), 3.3×10^{-16}	gauge, 2.2×10^{-16}
classifier, R_Z encoding	none	gauge, 2.2×10^{-16}
classifier, generic axis	none	gauge, 3.3×10^{-16}
classifier, noise inside the ring	none	none
classifier, encoding before first trainable layer	none	none
single-qubit regression, $f = \langle Z \rangle$	none	none
with a learnable sign	gauge, $\eta = -1$, 1.1×10^{-16}	gauge, $\eta = -1$, 2.8×10^{-16}
eigensolver, noise after each R_{ZZ}	gauge, 2.8×10^{-16}	gauge, 4.4×10^{-16}
eigensolver, noise after each CNOT	none	none

the preceding trainable blocks; $\Theta|0\rangle$ is absorbed by the first block, and the sign of the features by the readout. Proposition 1 therefore applies for every encoding axis (Table III). When the noise acts after each CNOT of the ring, no frame of either type exists; the exhaustive search of Table III rules out both.

In the classifier with the raw readout, we find that AD^{flip} reaches 0.838 ± 0.008 against 0.841 ± 0.004 for AD at $L = 4$ (95% CI of the difference $[-0.008, +0.002]$), with 96.7% of the individual predictions in agreement. By Corollary 1, the two conditions are identically distributed over initializations, so the expected difference is exactly zero; predictions differ because paired runs start from the same θ under both channels, not from θ and θ' . The comparison therefore has a known null answer. The two-level interval includes zero (paired t test, $P = 0.21$), whereas an interval that resamples only the test images, $[-0.0056, -0.0009]$, excludes it: resampling the test set alone understates the uncertainty.

When the noise acts after each CNOT of the ring, no frame exists, and the two directions define different model classes. Trained with the standardized readout at $L = 4$, AD^{flip} reaches 0.867 ± 0.009 against 0.873 ± 0.004 for AD, CI $[-0.013, +0.001]$; the two agree on 94.6% of the test images and in sensitivity (0.639 and 0.647). This variant applies the channel more often, so we compare its two directions only with each other. Breaking the gauge therefore allows a direction effect but does not guarantee a large one; here the trainable layers and the readout may compensate for it, whereas the eigensolver, with a fixed observable, cannot (Sec. IV C).

C. Eigensolver

In the eigensolver, the global flip $X^{\otimes 3}$ leaves H invariant. When the noise follows the complete R_{ZZ} , $X^{\otimes 2}$ commutes with the gate and the flip reaches the initial

R_Y rotations, where it becomes a reparametrization,

$$E_{\text{flip}}(\theta) = E_{\text{AD}}(\theta'), \quad \theta'_q = \pi - \theta_q, \quad (43)$$

where θ_q is the initial R_Y angle on qubit q and all other angles are unchanged. We find that the two sides of Eq. (43) agree to 1.4×10^{-15} over 50 random parameter sets at every γ . Trajectories started from mapped initial parameters end at energies that agree to 6×10^{-15} , and the optimized AD and reversed-AD energies coincide [Fig. 4(a)]. The initialization of Ref. [22] is also invariant under this map, but its optimizer, COBYLA [44], is not (Sec. IV A).

When the noise acts after each CNOT, the frame leaves a Pauli X on the target qubit between two noise channels, and the map fails: the energies of the two channels at mapped parameters differ by 0.10 to 0.66 between $\gamma = 0.01$ and 0.2. The optimized energies separate [Fig. 4(b)]. The median relative error of reversed AD exceeds that of AD by 0.06, 0.38, 0.56, and 0.65 at $\gamma = 0.01, 0.05, 0.1$, and 0.2 (Appendix C); small initial angles, drawn from $\mathcal{N}(0, 0.1^2)$, give the same gaps within 0.01 at $\gamma \geq 0.05$. The best of the 50 initializations for reversed AD lies within 0.01 of its median and above the median for AD at every γ . L-BFGS-B [45] from six further random initializations, run on an independently written simulator, reaches the same minima at $\gamma = 0.2$ (relative errors 0.0370 for AD and 0.6881 for reversed AD). The gap is therefore a floor of the energy landscape, not a failure of the optimizer. This reproduces the direction effect of Ref. [22] and locates its origin in the placement of the noise inside the decomposed gate. On hardware whose native two-qubit gate is the CNOT, the damping acts at this location.

On a bipartite coupling graph the effect cannot depend on the sign of the coupling. Flipping the spins of one sublattice, $\prod_{i \in B} X_i$, maps J to $-J$ and commutes with the transverse field; it acts after the last channel,

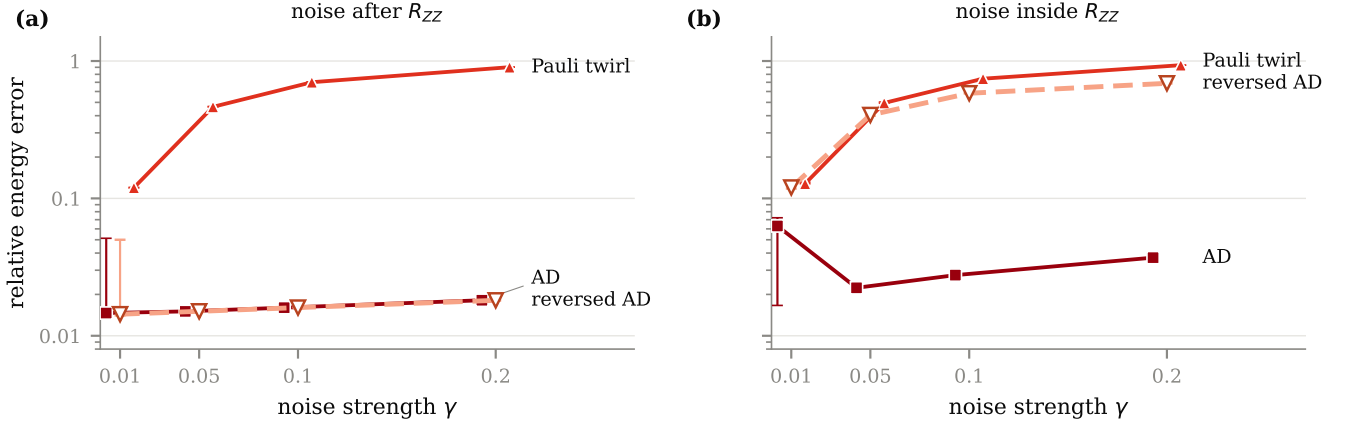


FIG. 4. Relative energy error of the three-qubit eigensolver against the noise strength, for AD, reversed AD (hollow markers), and the Pauli twirl (logarithmic vertical axis). Points are the median of 50 initializations and bars the interquartile range. (a) The channel acts after the complete R_{ZZ} ; AD and reversed AD coincide. (b) The channel acts after each CNOT inside R_{ZZ} , as in Ref. [22].

where the final R_X layer absorbs it ($x_i \mapsto x_i + \pi$ for $i \in B$). On the open chain, with B the middle qubit, the energies of the two couplings agree at the mapped parameters to 2×10^{-15} for every channel and placement, and the shift preserves the uniform initialization. On the three-site ring no such frame exists, since no product of Pauli operators flips every bond of an odd cycle; we have not run the ring with $J < 0$. With $J = \pm 1$, $h = 0.5$, and $\gamma = 0.1$ on the open chain, reversed AD reaches a median relative error of 0.367 against 0.063 for AD in both cases, while the bond correlation $\langle Z_i Z_{i+1} \rangle$ of the optimized AD state is +0.89 and -0.89. At $h = 2$ on the ring, where the ground state is paramagnetic, reversed AD still trails AD with the damping inside the gate (0.59 against 0.29 at $\gamma = 0.1$), and the two coincide with the damping after it. Which direction the trapped frame favors is thus set by the circuit, not by the ground state. Nor does the advantage of AD over its twirl rest on the proximity of the damping's fixed point $|000\rangle$ to the ground state: at $h = 2$ on the ring, $|000\rangle$ and every state the final R_X layer makes of it have a relative error of at least 0.54, yet AD reaches 0.29 and 0.23 with the damping inside and after the gate, against 0.66 and 0.59 for the twirl. (The antiferromagnetic open chain does not test this: the sublattice flip maps it onto the ferromagnetic chain, where $|000\rangle$ lies at 0.17.) On the ring, in both placements, the Pauli twirl gives the largest error, 0.70 and 0.74 at $\gamma = 0.1$. Unlike a classifier output, the energy cannot be rescaled freely, so here the twirl costs accuracy unless a mitigation step restores the scale [24].

V. WHAT THE NON-UNITAL TERM CHANGES

The output scale of Sec. III is not independent of the non-unital term. The Pauli twirl has exactly the contraction matrix of AD and differs only by $\mathbf{t} = \mathbf{0}$, so the larger feature scale under AD is itself an effect of $\mathbf{t}$. Here we show how $\mathbf{t}$ sets that scale and the sampling cost that follows from it (Sec. V A), and what it changes beyond them.

A. Sampling cost

Before training, the feature scale separates the channels with depth and width (Fig. 5). The unital channels shrink it by a nearly constant factor per layer at every width. Under AD it settles after two or three layers on a floor of 0.067 to 0.075 for $n = 4$ to 8. In the exact average over parameters (Appendix D), the ratio of the AD and Pauli-twirl scales grows by a factor of 3.3 per layer at $n = 4$ and 4.5 for $n \geq 6$, so its square, which sets the relative shot cost, grows by 11 and 20 per layer. At $n = 4$ the squared ratio is 2.1 at $L = 1$ and 1.3×10^3 at $L = 4$. At $p = 0.3$ the floor has a closed form. The damping after the last entangler branches the Z component of each feature into the identity, which reaches the readout whatever the earlier layers do, and contributes

$$\delta z_j = p(1-p)[(1-p)r_z \cos a_j - c r_y \sin a_j] \quad (44)$$

to the feature, where (r_x, r_y, r_z) is the direction into which the final trainable block rotates Z_j . The noiseless scale falls with width, as expected for a scrambling circuit, but the floor changes little, so at $n = 8$ and $L \geq 4$ the AD features are as large as the noiseless ones. This

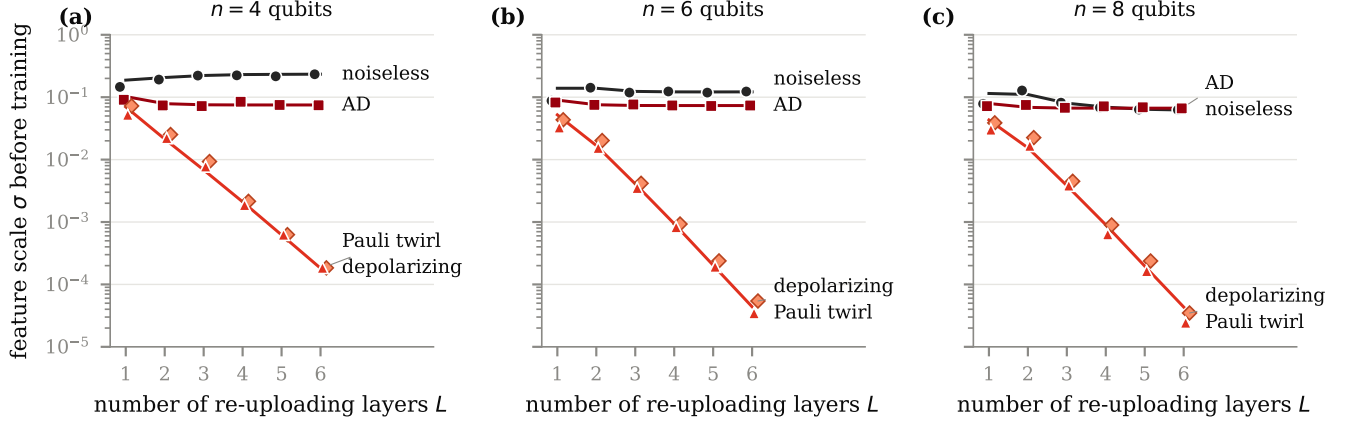


FIG. 5. Feature scale σ before training against the number of re-uploading layers for (a) four, (b) six, and (c) eight qubits, with n principal components as inputs. Markers are the root-mean-square scale over five parameter draws, the same at every depth, and 300 training images; lines are the exact average over parameters for the noiseless circuit, AD, and the Pauli twirl (Appendix D).

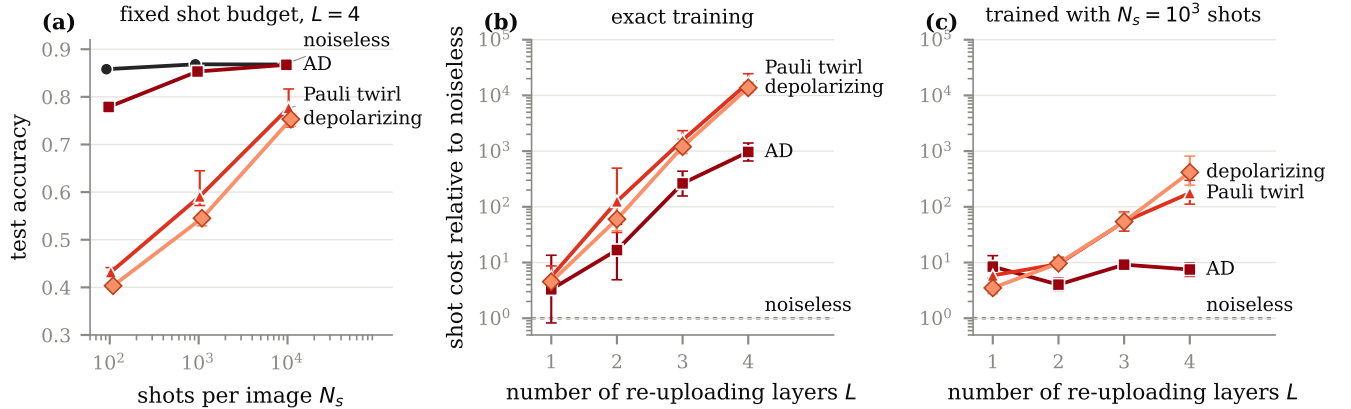


FIG. 6. Finite shots in the MNIST classifier. (a) Test accuracy at $L = 4$ of models trained and tested with N_s shots per image (median and interquartile range over eight initializations). (b), (c) Operational shot cost relative to the noiseless circuit (geometric mean over initializations, with 95% bootstrap intervals over initializations) for models trained with exact expectation values and with $N_s = 10^3$ shots.

is consistent with the effective shallowness of noisy circuits [11]; under AD the last layer alone sets the scale of the features. It also agrees with the theory of noise-induced concentration: unital noise concentrates local expectation values exponentially in depth [7, 13], whereas under non-unital noise the variance of local expectation values over random circuits stays bounded below by the size of the non-unital term [11]. Equation (D2) is the counterpart of that bound for the variance over inputs: branchings in different layers are uncorrelated after the average over parameters, so $\sigma_{\text{AD}}^2 \geq \sigma_{\infty}^2$ at every depth. Both bounds scale as $\|\mathbf{t}\|^2 = p^2$ and do not decay with depth; at weak damping the branchings of earlier layers raise the floor above this bound. The same effective shallowness allows local expectation values of such circuits to be estimated classically on average [11], and for

Pauli noise also by truncating Pauli paths [46]. For quantum kernels, noise likewise concentrates the values over different inputs, so that polynomially many shots give a model independent of the input [47]. In reservoir computing, the state of a contractive reservoir is the maximally mixed state for every input sequence if and only if the reservoir map is unital [48].

At finite temperature the damping of a qubit with splitting $\hbar\omega$ relaxes toward the thermal population $N = (e^{\hbar\omega/k_B T} + 1)^{-1}$, and at fixed p , that is, at fixed measured T_1 , $t_z = p(1 - 2N) = p \tanh(\hbar\omega/2k_B T)$. At fixed coupling to the bath, $1/T_1$ itself grows as $\coth(\hbar\omega/2k_B T)$, so at weak damping t_z is independent of temperature to first order in p , and temperature acts through the contraction. The last-layer term, Eq. (44), is linear in t_z , so it scales exactly as $1 - 2N$. Strings of higher Pauli

weight can branch into the identity on several qubits and add higher powers of t_z , but they are small: in the exact average over parameters at $n = 4$, the branching part of the feature scale follows $|1 - 2N|$ within 0.4% for every N and $L \leq 8$ at $p = 0.3$ (within 0.2% at $p = 0.1$), and it is exactly symmetric under $N \leftrightarrow 1 - N$, as the frame requires. The floor thus vanishes linearly in $1 - 2N$ as the temperature rises, and the Pauli twirl is the infinite-temperature end of this family.

After training with exact expectation values, the measured shot costs follow κ for the unital channels but not for AD [Fig. 6(b)]. At $L = 4$ on MNIST the operational cost relative to the noiseless circuit is 1.7×10^4 for the Pauli twirl and 1.4×10^4 for depolarizing noise, against $\kappa = 1.6 \times 10^4$ and 1.1×10^4 , and 967 for AD, against $\kappa = 50$. The Pauli twirl then needs 17 times the shots of AD on MNIST and 36 times on Fashion-MNIST. The excess of AD comes from the readout. The shot factor averages the spreads of the features, whereas the readout amplifies the shot noise of each feature by its own weight. The readout-weighted noise gain $G = \langle \sum_j \|W_{:,j}\|^2 (1 - z_j^2) / \sigma_j^2 \rangle$, averaged over the test images, reproduces the measured ratios to within a factor of 1.6 for $L \geq 2$. Exact training lets the AD readout rely on features whose spread is small: at $L = 4$ the geometric mean of the smallest feature spread is 0.0065 for AD, against 0.24 for the noiseless circuit.

Training under shot noise changes this picture [Fig. 6(a) and (c)]. With N_s shots in training and test, AD stays on average within 1.5 percentage points of the noiseless accuracy for $N_s \geq 10^3$ at every depth, whereas at $N_s = 10^3$ and $L = 4$ the Pauli and depolarizing channels lose 27 and 33 points, and at $N_s = 10^4$ still 9 and 12 points. At $N_s = 10^2$ AD also loses 8 points. Part of the loss arises in training. Evaluated with exact expectation values, the twirled models trained with 10^3 shots reach 0.762 and 0.758, against 0.870 when trained exactly, while AD reaches 0.864, against 0.876. The operational cost relative to the noiseless circuit then stays between 4 and 10 for AD at every depth, close to its shot factor (7.5 against 8.5 at $L = 4$). For the twirls it still grows by a factor of 3 to 5 per layer, to 177 and 417 at $L = 4$. Training under shot noise moves the circuit toward features whose spread exceeds the shot noise. At $L = 4$ and $N_s = 10^3$ the geometric mean of the smallest feature spread is 0.075 for AD, against 0.0065 under exact training, above $1/\sqrt{N_s} \approx 0.03$ and at the floor of Fig. 5; for the twirls it stays at 0.005 and 0.003, below the shot noise. At $L = 1$ the cost of AD is the largest of the three channels (8.4, against 5.9 and 3.5), with all costs small; we do not have an explanation for this.

The separation depends on the damping strength (Table IV and Fig. 7). At weak damping the twirled scale decays per layer as e^{-np} , and the square of the AD floor has two parts, $\sigma_{\text{br},\infty}^2 \approx \sigma_\infty^2 + b_{\text{br}} p$: the last-layer term of Eq. (D2), which is proportional to p^2 , and the branchings of the roughly $1/(np)$ earlier layers that the damping does not yet suppress. The second part acts on the data

TABLE IV. Dependence on the damping strength p before training, from the exact average over parameters: the depth L^* at which $(\sigma_{\text{AD}}/\sigma_{\text{Pauli}})^2$ first reaches 10, the floor $\sigma_{\text{br},\infty}$ of the AD scale and its last-layer part σ_∞ [Eq. (D2)], and the number of shots $1/\sigma_{\text{br},\infty}^2$ that resolves the floor. The floor is estimated from 5000 ($n = 4$) and 700 ($n = 8$) random image pairs, with a sampling error of about 1%.

p	L^*	$n = 4$			$n = 8$
		$\sigma_{\text{br},\infty}$	σ_∞	$1/\sigma_{\text{br},\infty}^2$	L^*
0.001	1508	0.0017	0.0005	3.4×10^5	–
0.003	453	0.0031	0.0014	1.0×10^5	–
0.01	117	0.0067	0.0047	2.2×10^4	49
0.03	33	0.0155	0.0137	4.2×10^3	13
0.05	18	0.0235	0.0219	1.8×10^3	8
0.1	8	0.0408	0.0396	600	5
0.2	4	0.0647	0.0637	240	3
0.3	2	0.0757	0.0747	170	2

dependence of the earlier layers, which concentrates with width, so that for $p \leq 0.03$ the coefficient b_{br} falls from $(1.8\text{--}2.7) \times 10^{-3}$ at $n = 4$ to $(1.0\text{--}2.4) \times 10^{-4}$ at $n = 8$. The squared ratio of the AD and twirl scales, which is the ratio of their shot costs, reaches 10 at the depth

$$L^* \approx \frac{1}{np} \ln \frac{3\sigma_{\text{None}}}{\sigma_{\text{br},\infty}}, \quad (45)$$

where σ_{None} is the noiseless scale at large depth (0.24 at $n = 4$ and 0.062 at $n = 8$). At $n = 4$ this agrees [Fig. 7(c)] with the exact value within 1% for $p \leq 0.01$ and within one layer for larger p , and at $n = 8$ within two layers for $p \leq 0.03$; it grows roughly as $p^{-1} \ln(1/p)$. At fixed $p \leq 0.05$ it falls faster than $1/n$ (from 117 at $n = 4$ to 49 at $n = 8$ for $p = 0.01$), since the noiseless scale in the logarithm concentrates with width faster than the floor. Damping on superconducting hardware is of order 10^{-3} per two-qubit gate, for gate times of 30 to 400 ns against T_1 of order 100 μs [9]; in our model one of the two insertions per layer follows a ring of n CNOTs, so p per insertion corresponds to several gate durations. At $p = 10^{-3}$ the separation needs about 1500 layers at $n = 4$, and resolving the AD floor needs 3×10^5 shots. The difference in feature scale between AD and its twirls is thus a property of strong damping or of very deep circuits. Trained classifiers follow this picture (Table V). At $L = 4$ and $p = 0.05$, well below $L^* = 18$, AD and its twirls reach the same accuracy at nearly the same shot cost, in agreement with κ . At $p = 0.1$, closer to $L^* = 8$, the Pauli twirl needs 1.6 times the shots of AD, between the ratio 1.4 of the trained feature scales and the squared ratio 1.7 before training, and AD is more accurate than the Pauli twirl by 0.009, CI [+0.002, +0.016]. At $p = 0.1$ and $L = L^* = 8$ the channels separate: the twirl needs 4.1 times the shots of AD, less than the ratios 7.8 of the trained and 16.8 of the untrained scales but in the same direction, and AD is more accurate than the Pauli twirl

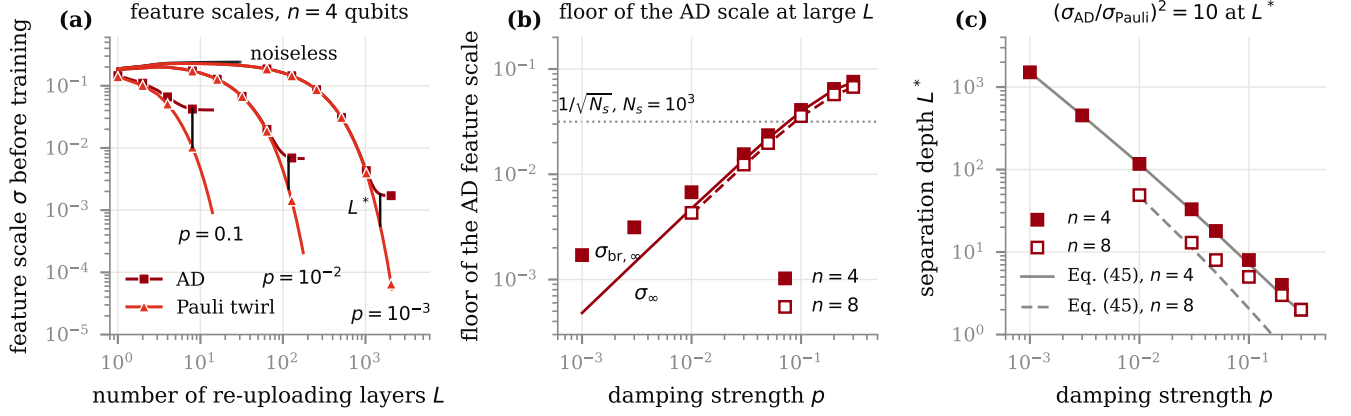


FIG. 7. Dependence of the separation on the damping strength, from the exact average over parameters before training (5000 image pairs at $n=4$, 700 at $n=8$). (a) Feature scale against depth at $n=4$ for AD (squares) and the Pauli twirl (triangles) at three values of p , and for the noiseless circuit up to $L=30$; vertical segments mark L^* . (b) Floor $\sigma_{br,\infty}$ of the AD scale (markers) and its last-layer part σ_∞ [Eq. (D2), lines] for $n=4$ (filled, solid) and $n=8$ (hollow, dashed); the dotted line is the shot noise $1/\sqrt{N_s}$ at $N_s=10^3$. (c) Separation depth L^* (markers) and Eq. (45) (gray lines). At $n=8$ the exact average was computed for $p \geq 0.01$.

by 0.026, CI $[+0.017, +0.035]$. This difference survives the standardized readout, so it is not one of scale; AD also fits the training set better (0.921 against 0.908). At this depth the noiseless circuit generalizes less well within the 100 training steps (training accuracy 0.910), and AD exceeds it by 0.035, CI $[+0.024, +0.047]$, while the twirls do not differ from it resolvably. A third of that difference is a better fit and two thirds a smaller generalization gap (0.054 against 0.079), with a lower input sensitivity ($\chi = 1.28$ against 2.14). AD does not exceed the best noiseless circuit (0.869 at $L=2$ to 4), and the gain needs many shots: evaluated with 10^3 shots per image, AD reaches 0.69 against 0.83 for the noiseless circuit, and it is ahead only at 10^4 shots and above. We report it as an observation at a fixed training budget and eight initializations; we have not tested whether it persists with longer training or more data.

The rescaling is cheap compared with generic error mitigation. Probabilistic error cancellation of the twirled channel needs a quasi-probability norm $\Gamma_{PEC} = 1/c^2$ per insertion, 1.43 at $p=0.3$. Over the 36 insertions of the $L=4$ classifier, the number of shots grows by

$$\Gamma_{PEC}^{2 \times 36} = (1-p)^{-72} \approx 10^{11}, \quad (46)$$

consistent with the exponential lower bounds on mitigation [10, 49]. A rescaling learned under shot noise costs far less: relative to the noiseless circuit, a factor of 177 to 417 in shots for the unital channels and 7.5 for AD at $L=4$ and $N_s=10^3$. It restores only the scale of the few features the task uses, not the noiseless state.

B. Accuracy and input sensitivity

With the output scale removed, AD ends slightly above the noiseless circuit at three and four layers on MNIST and at four layers on Fashion-MNIST (Sec. III C). The Pauli twirl isolates the non-unital term. Against it, AD is higher by 0.007 on MNIST at $L=4$, CI $[-0.001, +0.015]$, and by 0.013 on Fashion-MNIST, CI $[+0.003, +0.023]$. The gain is small; against the Pauli twirl it is resolved only on Fashion-MNIST, and we do not see it at one or two layers. It rests on eight initializations, and it is a result for exact expectation values: trained and tested with 10^3 shots per image, AD is 0.7 to 1.5 points below the noiseless circuit at $L=1$ to 4 (Sec. V A). At $p=0.1$ and $L=8$ the gain over the noiseless circuit is larger, 0.035, where the noiseless circuit generalizes less well (Sec. V A), and the sensitivity χ is again lower under AD, 1.28 against 2.14 for the noiseless circuit and 1.95 for the Pauli twirl. Gains from noise have been reported for reservoir computers under amplitude damping [50, 51] and for the generalization of variational models [52–54]; in quantum kernel machines, decoherence acts as an implicit regularization, although the test error there grows with the dephasing rate [55].

On MNIST the non-unital term also changes the input dependence of the trained classifier. Figure 3(c) shows the input sensitivity against depth. For the noiseless circuit it grows from 0.65 at $L=1$ to 1.00 at $L=4$, as each re-uploading layer adds to the dependence of the output on the input. The unital channels follow this growth, reaching 0.94 at $L=4$. Under AD the sensitivity barely grows, from 0.64 to 0.76. The difference from the noiseless circuit is -0.006 at $L=1$, within the spread over initializations, and grows to -0.07 , -0.16 , and -0.25 at $L=2, 3$, and 4, each well outside it

TABLE V. Trained classifiers at weaker damping (MNIST, standardized readout, exact expectation values, eight initializations): test accuracy, operational shot cost relative to the noiseless circuit at the same depth, shot factor κ , and the squared ratio of the AD and Pauli-twirl scales before training, $(\sigma_{\text{AD}}/\sigma_{\text{Pauli}})^2$, from the exact average over parameters (Sec. V A). The row $p = 0.3$ repeats Secs. III C and V A.

p	L	test accuracy				shot cost / noiseless			κ			before training
		None	AD	Pauli	Depol	AD	Pauli	Depol	AD	Pauli	Depol	
0.3	4	0.869	0.876	0.870	0.870	967	1.7×10^4	1.4×10^4	50	1.6×10^4	1.1×10^4	1.3×10^3
0.05	4	0.869	0.866	0.866	0.864	4.7	4.8	4.1	4.6	4.7	4.4	1.06
0.1	4	0.869	0.872	0.863	0.863	13.1	21.5	14.9	15.6	21.2	17.5	1.69
0.1	8	0.831	0.867	0.840	0.838	115	466	495	62	484	459	16.8

(Appendix C). Against the Pauli twirl, which isolates the non-unital term, the difference is -0.03 , -0.11 , and -0.18 at $L = 2, 3$, and 4 , the latter two well resolved (paired t test, $P = 0.007$ and 7×10^{-4} , unadjusted). For $L \geq 2$ the twirls themselves lie slightly below the noiseless circuit, by at most 0.06 , not resolved after correction for multiple tests (Appendix C). The sensitivity also depends on the gain of the readout, but the norm of the readout weights is similar for all channels (4.4 to 4.7 at $L = 4$), so the difference lies in the features. The sensitivity of the standardized features themselves, $\langle \|\partial \tilde{\mathbf{z}} / \partial \mathbf{a}\| \rangle$ with $\tilde{\mathbf{z}} = \text{diag}(\boldsymbol{\sigma})^{-1}(\mathbf{z} - \boldsymbol{\mu})$, confirms this. At $L = 4$ it is 2.79 for AD, against 4.13 for the noiseless circuit and 3.98 for the Pauli twirl ($P = 10^{-5}$ against the twirl, paired t test); on Fashion-MNIST no channel differs from another. The lower sensitivity makes the MNIST classifier more robust to a miscalibrated encoding: a common offset of ± 1 on the four test angles at $L = 4$ costs AD 0.29 to 0.33 in accuracy, against 0.44 to 0.55 for the other channels. Depolarizing noise provably limits the effect of small adversarial perturbations of the input [56]. The offset here is large, and at equal contraction only the non-unital channel reduces the drop.

On Fashion-MNIST at $L = 4$, however, we find no such difference. The sensitivities χ are 2.56 for the noiseless circuit, 2.86 for AD, 2.89 for the Pauli twirl, and 3.16 for depolarizing noise, none resolved from the noiseless value, and AD is not more robust to an offset. The effect on the input dependence is therefore specific to the data. A candidate explanation is the effective shallowness of noisy circuits [11]. It holds for unital and non-unital noise alike, but what survives the truncation differs: under the twirls the features contain only contributions that survive the contraction of every layer, whereas under AD they also carry the last-layer term of Eq. (44), a first harmonic of a single angle, which dominates at depth before training. We have not tested whether this term accounts for the lower sensitivity on MNIST and its absence on Fashion-MNIST.

VI. CONCLUSION

In this work, we have compared amplitude damping with its twirls, and in particular with the Pauli twirl, which is damping at infinite temperature. The zero-temperature bias acts on variational models mainly through the scale of their features. For random parameters, the features under AD settle on a floor, set at strong damping by the last layer and falling linearly in $1 - 2N$ with temperature at fixed T_1 , while the twirls shrink every feature by a fixed factor per layer, at every width up to eight qubits. A trainable output scale removes most of the resulting differences in fit and accuracy in exact simulation, in the single-qubit fits of Ref. [22] and in a four-qubit classifier on MNIST and Fashion-MNIST; the rest reappears as a sampling cost. Trained and tested with 10^3 shots per image at $p = 0.3$, the classifier under AD stays within 1.5 percentage points of the noiseless accuracy at depths 1 to 4 , whereas the twirled classifiers lose up to 33 points at four layers. The separation needs a depth that grows roughly as $(np)^{-1} \ln(1/p)$, about 1500 layers for four qubits at $p = 10^{-3}$ and fewer at larger width, so it is a property of strong damping or of very deep circuits.

Reversing the damping direction is a reparametrization whenever a Pauli frame or spin time reversal can reach trainable gates and the readout; with a symmetric initialization and an equivariant optimizer, trained models under the two directions are then identically distributed, as in our classifier. The direction matters only where the circuit traps the frame, as in the eigensolver with the damping inside the decomposed two-qubit gate; there reversed damping does worse also when the transverse field dominates, and on an open chain the sign of the coupling is itself a gauge. Beyond the scale, the bias leaves AD ahead of its twirl by up to three percentage points in exact simulation, most at the separation depth among the depths we trained (2.6 points at $p = 0.1$, $L = 8$), and it lowers the input sensitivity on MNIST only.

These results suggest three practices for comparisons of noise structure in variational models. The first is to report the output scale, or to include a trainable one, together with the shot budget it implies. The second is to check whether the model class is invariant under

the relevant frames before attributing an effect to the direction of the noise. The third is to state where the noise acts relative to the decomposition of the gates.

Several questions remain open. Our simulations are exact and small, with one qubit for the fits, four for the classifier, and three for the eigensolver. The width dependence is established before training, up to eight qubits; training beyond four qubits was out of reach of our exact simulation, and our shot model in training neglects the correlations between qubits. The origin of the sensitivity effect on MNIST, and its absence on Fashion-MNIST, remain unexplained; measuring the contribution of each layer to the input dependence would be a first step. It also remains to be seen whether the effect persists for more qubits and other data, whether trained models follow the temperature dependence of the floor, and whether it survives on hardware, where damping comes with dephasing. Finally, Refs. [57, 58] studied the opposite question to ours: whether noise breaks a symmetry that a model is built to respect [59, 60]. The gauge freedom studied here is instead a symmetry of the circuit that makes the direction of the noise invisible.

ACKNOWLEDGMENTS

Claude Opus 5.5 (Anthropic) was used throughout the preparation of this work: to search and survey the literature, and to check its numerical claims and its characterizations of prior work against the underlying data and sources.

DATA AVAILABILITY

The code, the result files, and the scripts that reproduce every number and figure in this paper are available at <https://github.com/Identity-Operator/zero-vs-infinite-T-damping> and archived on Zenodo at [doi:10.5281/zenodo.23053251](https://doi.org/10.5281/zenodo.23053251) [61].

Appendix A: Reading of the protocol of Ref. [22]

We simulate the model of Ref. [22] (their App. B1) in the Bloch representation. Each of the $L+1 = 3$ trainable blocks $\mathcal{W}_l = \mathcal{U}(W_l)$ of Eq. (9) applies

$$W_l = R_Z(\theta_{3l}) R_Y(\theta_{3l+1}) R_Z(\theta_{3l+2}), \quad l = 0, 1, 2, \quad (\text{A1})$$

and then the channel; after each of the first two blocks follow $R_X(x)$ and the channel again. The output is $f(x) = r_z$. The channels act by Eq. (2) with the entries of Table I.

Where the description in Ref. [22] leaves a choice, we made the following one. The 250 inputs lie on a uniform grid over $[-2\pi, 2\pi]$. Each target g' of Eq. (11) has coefficients drawn uniformly from $[-1, 1]$ and is shifted to zero

mid-range, normalized to unit range, and multiplied by the range R ,

$$g(x) = R \frac{g'(x) - \frac{1}{2}(g'_{\max} + g'_{\min})}{g'_{\max} - g'_{\min}}, \quad (\text{A2})$$

where $g'_{\max}$ and $g'_{\min}$ are the extreme values on the grid. A step is one Adam update on a batch of 25 inputs; the batches follow a seeded shuffle of the 250 inputs, renewed every ten steps and shared by all channels. We use their settings for $L = 2$ (learning rate 0.20, $\beta_1 = 0.45$, $\beta_2 = 0.95$, 150 steps). Initial parameters are drawn uniformly from $[0, 2\pi]$, the same for every channel. A target has converged at the first step at which the mean squared error over all 250 inputs falls below threshold,

$$\frac{1}{250} \sum_{i=1}^{250} [f(x_i) - g(x_i)]^2 < 10^{-5}. \quad (\text{A3})$$

The largest converging range is found by bisection on $[0, 2]$ with ten halvings.

The raw model uses $f = \langle Z \rangle$ with the target centered on zero, or $f = \langle Z \rangle + b$ with a trainable b , which is equivalent to the free target offset b of their Eq. (7). The rescaled model is Eq. (12), with s and b trainable from $s = 1$ and $b = 0$, and is trained for up to 1000 steps at fixed R . The scale and offset are updated with the same Adam settings as the circuit parameters.

Appendix B: Proof of the frame gauge

a. Setting. A model on n qubits prepares $\rho_0 = |0\rangle\langle 0|^{\otimes n}$, applies elements $E_1, \dots, E_m$, and reads out

$$y_i = \text{Tr}[O_i \Phi(\rho_0)], \quad \Phi = E_m \circ \dots \circ E_1, \quad (\text{B1})$$

possibly followed by a learned affine map. Each E_k is a fixed unitary $\mathcal{U}_G$, a data encoding $\mathcal{U}_{S(x)}$ with $S(x) = \bigotimes_q R_{\mathbf{n}_q}(x_q)$, a trainable unitary from a family $\mathcal{W}$ (general single-qubit blocks, or Pauli rotations $e^{-i\theta Q/2}$), or a layer of the single-qubit channel AD on a set S_k of qubits. The reversed model replaces every noise layer by $\text{AD}^{\text{flip}} = \mathcal{X} \text{AD} \mathcal{X}$.

b. Frames. Let $\mathcal{P}(\rho) = P\rho P$ for a Pauli string P , and $\mathcal{K}(\rho) = \bar{\rho}$. On Hermitian operators $\mathcal{K}$ is real-linear, trace preserving and positive, $\mathcal{K}^2 = \text{id}$, and for any channel $\mathcal{E}$ with Kraus operators $\{K_j\}$ one has $\mathcal{K} \circ \mathcal{E} \circ \mathcal{K} = \bar{\mathcal{E}}$, the channel with Kraus operators $\{\bar{K}_j\}$. A frame of unitary type is $g = \mathcal{P}$; of antiunitary type, $g = \mathcal{P} \circ \mathcal{K}$. In both cases $g \circ g = \text{id}$, and

$$g \circ \mathcal{E} \circ g': \{K_j\} \mapsto \{P K_j^{(t)} P'\}, \quad (\text{B2})$$

with $K^{(t)} = K$ for the unitary type and $K^{(t)} = \bar{K}$ for the antiunitary type.

TABLE VI. Standardized-readout classifier on MNIST: difference from the noiseless circuit in test accuracy (Δacc) and input sensitivity ($\Delta\chi$). P values from paired t tests over the eight initializations, Holm-adjusted over the 22 tests of the table.

L	channel	Δacc	95% CI	P	$\Delta\chi$ (P)
1	AD	+0.001	$[-0.003, +0.005]$	1	-0.006 (1)
1	Pauli	-0.059	$[-0.178, +0.001]$	1	+0.058 (1)
1	Depol	0	exact	–	0 (–)
2	AD	+0.003	$[-0.003, +0.009]$	1	-0.072 (0.004)
2	Pauli	+0.004	$[-0.003, +0.010]$	1	-0.043 (0.26)
2	Depol	+0.000	$[-0.004, +0.004]$	1	-0.023 (1)
3	AD	+0.009	$[+0.003, +0.015]$	0.04	-0.162 (0.02)
3	Pauli	+0.001	$[-0.007, +0.008]$	1	-0.049 (1)
3	Depol	+0.000	$[-0.007, +0.007]$	1	-0.037 (1)
4	AD	+0.008	$[+0.002, +0.013]$	0.008	-0.247 (10^{-5})
4	Pauli	+0.001	$[-0.007, +0.009]$	1	-0.063 (0.57)
4	Depol	+0.001	$[-0.006, +0.008]$	1	-0.062 (1)

c. Telescoping. For any frames $g_0, \dots, g_m$ of one type,

$$\Phi^{\text{rev}} = g_m \circ (g_m E_m^{\text{rev}} g_{m-1}) \circ \dots \circ (g_1 E_1^{\text{rev}} g_0) \circ g_0, \quad (\text{B3})$$

which is an identity since every inserted $g_k g_k$ is the identity.

d. The conditions. (i) Consider a noise layer on S_k with $g_k = g_{k-1}$ acting as X or Y on each $q \in S_k$. The Kraus operators of AD are real, and $ZADZ = \text{AD}$, so

$$XADX = YADY = \text{AD}^{\text{flip}}, \quad (\text{B4})$$

$$g_k \circ \text{AD}^{\text{flip}} \circ g_{k-1} = \text{AD} \quad (\text{B5})$$

for both types. Off S_k the frame passes unchanged.

(ii) For a fixed unitary G , $g_k \mathcal{U}_G g_{k-1} = \mathcal{U}_{P_k G^{(t)} P_{k-1}}$, which equals $\mathcal{U}_G$ if and only if

$$P_k \propto G P_{k-1} G^{(t)\dagger}; \quad (\text{B6})$$

the frame propagates by conjugation and must remain a Pauli string. For Clifford G this holds for every P_{k-1} of the unitary type; for real Clifford G (CNOT) also of the antiunitary type.

(iii) For an encoding, $g_k = g_{k-1}$ and $P_q R_{\mathbf{n}_q}^{(t)}(x) P_q = R_{\mathbf{n}_q}(x)$ for all x . For the unitary type this requires $[P_q, \mathbf{n}_q \cdot \boldsymbol{\sigma}] = 0$. For the antiunitary type, $\overline{R_{\mathbf{n}}(x)} = \exp[+ix \mathbf{n} \cdot \boldsymbol{\sigma}/2]$, so the condition is

$$P_q \overline{\mathbf{n}_q \cdot \boldsymbol{\sigma}} P_q = -\mathbf{n}_q \cdot \boldsymbol{\sigma}, \quad (\text{B7})$$

which holds for $P_q = Y$ and every $\mathbf{n}_q$, since $\bar{X} = X$, $\bar{Y} = -Y$, $\bar{Z} = Z$, $\bar{Y}\sigma_{x,z}Y = -\sigma_{x,z}$, and $YYY = Y$.

(iv) A general single-qubit block W is mapped to $P_{k,q} W^{(t)} P_{k-1,q} \in \text{SU}(2)$ for any pair of frames: such blocks absorb, and may change, the frame on their qubits. A Pauli rotation is mapped to

$$P e^{\mp i\theta Q^{(t)}/2} P = e^{-i\theta' Q/2}, \quad \theta' = \pm\theta, \quad (\text{B8})$$

if and only if $g_k = g_{k-1}$ and $PQ^{(t)}P = \pm Q$; it can carry a frame but not change it.

(v) By Eq. (B3) the input is $g_0(\rho_0) = P_0 \rho_0 P_0$ (as $\bar{\rho}_0 = \rho_0$). If the first element on qubit q is a general block W , the pair $W^{(t)} P_{0,q}$ is itself a general block acting on $|0\rangle$; if it is R_X or R_Y , $X|0\rangle$ and $Y|0\rangle$ equal $R_{X,Y}(\pi)|0\rangle$ up to phase and are absorbed by a shift of the angle; otherwise $P_{0,q}$ must fix $|0\rangle$, i.e. $P_{0,q} \in \{I, Z\}$. At the output, using $\text{Tr}[O\bar{\tau}] = \text{Tr}[\bar{O}\tau]$ for Hermitian arguments,

$$\text{Tr}[O g_m(\tau)] = \text{Tr}[P_m O^{(t)} P_m \tau], \quad (\text{B9})$$

for any state τ , which is $\eta \text{Tr}[O\tau]$ if and only if $P_m O^{(t)} P_m = \eta O$.

e. Conclusion. If (i)–(v) hold, every bracket in Eq. (B3) is an element of the original model with reparametrized trainable parameters, the input frame is absorbed by the first elements and the output frame contributes the sign η ; hence

$$y_i^{\text{rev}}(x; \theta) = \eta_i y_i(x; \theta') \quad (\text{B10})$$

for all x , which is Eq. (35). $\square$

f. Corollaries. The first statement of Corollary 1 follows because $\theta \mapsto \theta'$ is a bijection of the parameter space. For the second, write the map as $\theta' = S\theta + \pi\mathbf{m}$ with S a diagonal sign matrix and $\mathbf{m}$ an integer vector, as for every frame of Table III, together with $W \mapsto W \text{diag}(\boldsymbol{\eta})$ for the readout. The training loss then satisfies $\mathcal{L}^{\text{flip}}(S\theta + \pi\mathbf{m}, W \text{diag}(\boldsymbol{\eta}), b) = \mathcal{L}(\theta, W, b)$, so its gradients transform with S and $\text{diag}(\boldsymbol{\eta})$. An optimizer whose update commutes with these maps (gradient descent and Adam without weight decay exactly, SPSSA in distribution) carries the trajectory from (θ_0, W_0, b_0) under AD onto the trajectory from $(S\theta_0 + \pi\mathbf{m}, W_0 \text{diag}(\boldsymbol{\eta}), b_0)$ under AD^{flip} . If the map preserves the distribution of (θ_0, W_0, b_0) , the trained models are identically distributed. For Corollary 2, write a qubit channel as in Eq. (2). Unitary Pauli conjugation and

complex conjugation act on $(T, \mathbf{t})$ as

$$\mathcal{P}: (T, \mathbf{t}) \mapsto (DTD, D\mathbf{t}), \quad (\text{B11})$$

$$\mathcal{K}: (T, \mathbf{t}) \mapsto (FTF, F\mathbf{t}), \quad (\text{B12})$$

with D a diagonal sign matrix and $F = \text{diag}(1, -1, 1)$; both preserve $\|\mathbf{t}\|$. The Pauli twirl has $\mathbf{t} = 0$, while $\|\mathbf{t}\| = p$ for amplitude damping.

g. Time reversal and CNOT. By Eq. (42), time reversal leaves a local Pauli residue on the input side of each CNOT. A Z residue passes through amplitude damping ($ZADZ = AD$); an X residue does not. The residue must therefore reach a general trainable block without crossing a noise location, which is the case in our classifier (the trainable layer immediately precedes the ring) and fails when noise is inserted after each CNOT (Table III).

h. Scope. The proposition is a sufficient condition. The rows marked “none” in Table III establish that no frame of either type exists; they do not by themselves prove that the classes differ. For the single-qubit regression model the output range proves it. For the eigensolver at $\gamma = 0.3$, with L-BFGS-B from 12 random initializations and the channel applied once after each R_{ZZ} or after each CNOT, the lowest energies are equal where the gauge holds (-3.1770 for both) and differ by 1.98 where it does not (-3.087 against -1.109), reproducing the direction of the finding of Ref. [22]. For the classifier with noise inside the ring, Sec. IV B measures the difference directly.

Appendix C: Statistical tests

We collect here the hypothesis tests behind the comparisons in the main text. To avoid confusion with the noise strength p , we denote P values by a capital P . For the classifier, accuracy differences come with the 95% CI of the two-level bootstrap described in Sec. II (2×10^4 resamples) and the P value of the paired t test across the eight initializations, which the conditions share; sensitivity differences come with the paired t test. The P values of Table VI are Holm-adjusted over its 22 tests. Five remain below 0.05: the accuracy of AD at $L = 3$ and 4, and its sensitivity at $L = 2, 3$, and 4.

On Fashion-MNIST at $L = 4$, the differences from the noiseless circuit are $+0.017$, CI $[+0.007, +0.029]$, $P = 0.03$, for AD; $+0.005$, CI $[-0.004, +0.014]$, $P = 0.28$, for the Pauli twirl; and $+0.009$, CI $[-0.0001, +0.018]$, $P = 0.08$, for depolarizing noise (paired t , Holm-adjusted over the three channels). The sensitivity differences are $+0.30$ ($P = 0.25$), $+0.33$ ($P = 0.21$), and $+0.60$ ($P = 0.21$). AD minus the Pauli twirl in accuracy is -0.001 , CI $[-0.007, +0.005]$, at $L = 2$; $+0.008$, CI $[-0.0002, +0.017]$, at $L = 3$; $+0.007$, CI $[-0.001, +0.015]$, at $L = 4$ on MNIST; and $+0.013$, CI $[+0.003, +0.023]$, on Fashion-MNIST. At $L = 1$ on MNIST it is dominated by the diverged Pauli run.

For the classifier with the noise after each CNOT of the ring (Sec. IV B), AD^{flip} minus AD is -0.006 in accuracy,

CI $[-0.013, +0.001]$, $P = 0.10$, and -0.008 in sensitivity, $P = 0.71$. With the standard placement and the raw readout, AD^{flip} minus AD is -0.003 , CI $[-0.008, +0.002]$, $P = 0.21$.

For the single-qubit fits, convergence of AD and of the Pauli twirl on the same 200 targets is compared with McNemar’s exact test. Within 1000 steps, no difference is significant at either range ($P \geq 0.34$). Within 150 steps, the only significant difference is -13.5 percentage points at $p = 0.2$ and $R = 0.468$ ($P = 0.0013$); the others have $P \geq 0.077$. For the raw model, the largest converging ranges of AD and the Pauli twirl are compared with the paired Wilcoxon test: with a free target offset, $P = 0.71$, 0.049 , and 6×10^{-14} at $p = 0.05, 0.1$, and 0.2 ; with a centered target, $P = 0.19, 9 \times 10^{-7}$, and 0.04 .

For the eigensolver, the relative errors of reversed AD and AD over the 50 initializations are compared with the Mann–Whitney test. With the noise inside R_{ZZ} , $P < 10^{-16}$ at every γ ; with the noise after it, $P \geq 0.69$.

Appendix D: Feature scale before training

The lines in Fig. 5 are exact averages, over parameters drawn uniformly from $[0, 2\pi)$, of

$$\sigma^2 = \mathbb{E}_\theta \frac{1}{n} \sum_{j=1}^n \text{Var}_{\mathbf{a}} z_j, \quad (\text{D1})$$

where the variance runs over the first 300 training images. We compute them in the Heisenberg picture from the second moments of the Pauli-string coefficients of Z_j , with the parameters shared between two images. Every uniformly random $R_Z R_Y R_Z$ block removes the cross moments between different strings because, averaged over its angles, the entries R_{bc} of its adjoint action have vanishing first moments and different columns are uncorrelated, $\mathbb{E}_\theta[R_{xb}R_{yc}] = 0$ for $b \neq c$. Between two blocks the circuit acts site by site (channel, $R_X(a_j)$, channel), so the second moments of two images propagate through the element-wise product of the two composite site maps, followed by the entangling ring, which permutes the strings up to a sign. The second moments therefore obey a closed recursion, which is exact up to the sampling of image pairs: all pairs for $n \leq 5$, and 5000 random pairs for $n \geq 6$. Against brute-force averages of the density-matrix simulation, the recursion agrees within one to two standard errors in all 18 cells at $n = 2$ and 3 and $L = 1$ to 3. An independent density-matrix simulation at $n = 4$, with 200 parameter draws, agrees within two standard errors in all 12 cells for the noiseless circuit, AD, and the Pauli twirl at $L = 1$ to 4, and in all four cells for AD and the Pauli twirl at $p = 0.1$ and $L = 4$ and 8.

Under AD the adjoint channel maps Z to $c^2 Z + pI$. The next random block averages the cross moments between the I and Z branches to zero, so that $\sigma_{\text{AD}}^2 = \sigma_{\text{Pauli}}^2 + \sigma_{\text{br}}^2$, where σ_{br} collects the paths through at least one branching into the identity. The branching after the last

entangler gives Eq. (44). Averaged over the final block, with $\mathbb{E} r_z^2 = \frac{1}{2}$, $\mathbb{E} r_y^2 = \frac{1}{4}$, and $\mathbb{E} r_z r_y = 0$, it yields

$$\sigma_\infty^2 = \frac{p^2(1-p)^2}{n} \sum_j \left[\frac{(1-p)^2}{2} \text{Var}(\cos a_j) + \frac{1-p}{4} \text{Var}(\sin a_j) \right], \quad (\text{D2})$$

which at $p = 0.3$ is the floor of σ_{AD} within 0.5% for $n \geq 4$ (with all image pairs); branchings in earlier layers are at-

tenuated by the rest of the circuit. At weaker damping they are not, and they add the term $b_{\text{br}} p$ of Sec. V A. For uniformly distributed angles, $\sigma_\infty = 0.096$. The measured points in Fig. 5 use five parameter draws, the same parameter vectors at every depth (those of depth L are the leading entries of those of depth 6), so their deviations are correlated across L . The distribution over the parameters is heavy-tailed, and five draws tend to fall below the average, as for the Pauli twirl at $n = 8$ and $L \geq 4$. An independent Pauli-path simulation reproduces these five draws to 2×10^{-12} and, over 120 further draws at $L = 4$, agrees with the exact average within one standard error.

-
- [1] J. Preskill, Quantum computing in the NISQ era and beyond, *Quantum* **2**, 79 (2018), [arXiv:1801.00862](#).
 - [2] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles, Variational quantum algorithms, *Nature Reviews Physics* **3**, 625 (2021).
 - [3] K. Bharti, A. Cervera-Lierta, T. H. Kyaw, T. Haug, S. Alperin-Lea, A. Anand, M. Degroote, H. Heimonen, J. S. Kottmann, T. Menke, W.-K. Mok, S. Sim, L.-C. Kwek, and A. Aspuru-Guzik, Noisy intermediate-scale quantum algorithms, *Reviews of Modern Physics* **94**, 015004 (2022).
 - [4] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, Barren plateaus in quantum neural network training landscapes, *Nature Communications* **9**, 4812 (2018), [arXiv:1803.11173](#).
 - [5] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles, Cost function dependent barren plateaus in shallow parametrized quantum circuits, *Nature Communications* **12**, 1791 (2021), [arXiv:2001.00550](#).
 - [6] M. Larocca, S. Thanasilp, S. Wang, K. Sharma, J. Biamonte, P. J. Coles, L. Cincio, J. R. McClean, Z. Holmes, and M. Cerezo, Barren plateaus in variational quantum computing, *Nature Reviews Physics* **7**, 174 (2025), [arXiv:2405.00781](#).
 - [7] S. Wang, E. Fontana, M. Cerezo, K. Sharma, A. Sone, L. Cincio, and P. J. Coles, Noise-induced barren plateaus in variational quantum algorithms, *Nature Communications* **12**, 6961 (2021), [arXiv:2007.14384](#).
 - [8] D. Stilck França and R. García-Patrón, Limitations of optimization algorithms on noisy quantum devices, *Nature Physics* **17**, 1221 (2021), [arXiv:2009.05532](#).
 - [9] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson, and W. D. Oliver, A quantum engineer's guide to superconducting qubits, *Applied Physics Reviews* **6**, 021318 (2019), [arXiv:1904.06560](#).
 - [10] Y. Quek, D. Stilck França, S. Khatri, J. J. Meyer, and J. Eisert, Exponentially tighter bounds on limitations of quantum error mitigation, *Nature Physics* **20**, 1648 (2024).
 - [11] A. A. Mele, A. Angrisani, S. Ghosh, S. Khatri, J. Eisert, D. Stilck França, and Y. Quek, Noise-induced shallow circuits and absence of barren plateaus, *Nature Physics* **22**, 751 (2026), [arXiv:2403.13927](#).
 - [12] B. Fefferman, S. Ghosh, M. Gullans, K. Kuroiwa, and K. Sharma, Effect of nonunitary noise on random-circuit sampling, *PRX Quantum* **5**, 030317 (2024), [arXiv:2306.16659](#).
 - [13] M. Schumann, F. K. Wilhelm, and A. Ciani, Emergence of noise-induced barren plateaus in arbitrary layered noise models, *Quantum Science and Technology* **9**, 045019 (2024).
 - [14] P. Singkanipa and D. A. Lidar, Beyond unital noise in variational quantum algorithms: noise-induced barren plateaus and limit sets, *Quantum* **9**, 1617 (2025), [arXiv:2402.08721](#).
 - [15] S. Schmitt, L. Ekstrøm, A. Bottarelli, and X. Bonet-Monroig, Experimental demonstration of the absence of noise-induced barren plateaus using information content landscape analysis (2026), [arXiv:2602.22851](#).
 - [16] A. Sannia, F. Tacchino, I. Tavernelli, G. L. Giorgi, and R. Zambrini, Engineered dissipation to mitigate barren plateaus, *npj Quantum Information* **10**, 81 (2024), [arXiv:2310.15037](#).
 - [17] J. J. Wallman and J. Emerson, Noise tailoring for scalable quantum computation via randomized compiling, *Physical Review A* **94**, 052325 (2016).
 - [18] A. Hashim, R. K. Naik, A. Morvan, J.-L. Ville, B. Mitchell, J. M. Kreikebaum, M. Davis, E. Smith, C. Iancu, K. P. O'Brien, I. Hincks, J. J. Wallman, J. Emerson, and I. Siddiqi, Randomized compiling for scalable quantum computing on a noisy superconducting quantum processor, *Physical Review X* **11**, 041039 (2021).
 - [19] K. Temme, S. Bravyi, and J. M. Gambetta, Error mitigation for short-depth quantum circuits, *Physical Review Letters* **119**, 180509 (2017), [arXiv:1612.02058](#).
 - [20] Z. Cai, R. Babbush, S. C. Benjamin, S. Endo, W. J. Huggins, Y. Li, J. R. McClean, and T. E. O'Brien, Quantum error mitigation, *Reviews of Modern Physics* **95**, 045005 (2023), [arXiv:2210.00921](#).
 - [21] C. Dankert, R. Cleve, J. Emerson, and E. Livine, Exact and approximate unitary 2-designs and their application to fidelity estimation, *Physical Review A* **80**, 012304 (2009), [arXiv:quant-ph/0606161](#).
 - [22] C. van Rossum, S. Shrapnel, and R. Gupta, Exploiting biased noise in variational quantum models, *Quantum Science and Technology* **11**, 025047 (2026), [arXiv:2510.24050](#).
 - [23] S. Wang, P. Czarnik, A. Arrasmith, M. Cerezo, L. Cincio, and P. J. Coles, Can error mitigation improve trainability of noisy variational quantum algorithms?, *Quantum* **8**,

- 1287 (2024).
- [24] J. Vovrosh, K. E. Khosla, S. Greenaway, C. Self, M. S. Kim, and J. Knolle, Simple mitigation of global depolarizing errors in quantum simulations, *Physical Review E* **104**, 035309 (2021).
 - [25] E. Fontana, M. Cerezo, A. Arrasmith, I. Rungger, and P. J. Coles, Non-trivial symmetries in quantum landscapes and their resilience to quantum noise, *Quantum* **6**, 804 (2022), [arXiv:2011.08763](#).
 - [26] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th ed. (Cambridge University Press, 2010).
 - [27] W. Dür, M. Hein, J. I. Cirac, and H.-J. Briegel, Standard forms of noisy quantum operations via depolarization, *Physical Review A* **72**, 052326 (2005), [arXiv:quant-ph/0507134](#).
 - [28] M. R. Geller and Z. Zhou, Efficient error models for fault-tolerant architectures and the Pauli twirling approximation, *Physical Review A* **88**, 012314 (2013), [arXiv:1305.2021](#).
 - [29] M. Gutiérrez, L. Svec, A. Vargo, and K. R. Brown, Approximation of realistic errors by Clifford channels and Pauli measurements, *Physical Review A* **87**, 030302 (2013), [arXiv:1207.0046](#).
 - [30] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre, Data re-uploading for a universal quantum classifier, *Quantum* **4**, 226 (2020), [arXiv:1907.02085](#).
 - [31] M. Schuld, R. Sweke, and J. J. Meyer, The effect of data encoding on the expressive power of variational quantum machine learning models, *Physical Review A* **103**, 032430 (2021).
 - [32] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, Gradient-based learning applied to document recognition, *Proceedings of the IEEE* **86**, 2278 (1998).
 - [33] D. P. Kingma and J. Ba, Adam: A method for stochastic optimization (2014), [arXiv:1412.6980 \[cs.LG\]](#).
 - [34] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, PyTorch: An imperative style, high-performance deep learning library, in *Advances in Neural Information Processing Systems 32* (2019) [arXiv:1912.01703](#).
 - [35] H. Xiao, K. Rasul, and R. Vollgraf, Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms (2017), [arXiv:1708.07747](#).
 - [36] A. Javadi-Abhari, M. Treinish, K. Krsulich, C. J. Wood, J. Lishman, J. Gacon, S. Martiel, P. D. Nation, L. S. Bishop, A. W. Cross, B. R. Johnson, and J. M. Gambetta, Quantum computing with Qiskit (2024), [arXiv:2405.08810](#).
 - [37] C. van Rossum, S. Shrapnel, and R. Gupta, *Code base for exploiting biased noise in variational quantum models*, figshare, version 3 (2026).
 - [38] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap* (Chapman and Hall/CRC, New York, 1994).
 - [39] D. Wecker, M. B. Hastings, and M. Troyer, Progress towards practical quantum variational algorithms, *Physical Review A* **92**, 042303 (2015), [arXiv:1507.08969](#).
 - [40] J. F. Gonthier, M. D. Radin, C. Buda, E. J. Duskocil, C. M. Abuan, and J. Romero, Measurements as a roadblock to near-term practical quantum advantage in chemistry: Resource analysis, *Physical Review Research* **4**, 033154 (2022), [arXiv:2012.04001](#).
 - [41] M. Urbanek, B. Nachman, V. R. Pascuzzi, A. He, C. W. Bauer, and W. A. de Jong, Mitigating depolarizing noise on quantum computers with noise-estimation circuits, *Physical Review Letters* **127**, 270502 (2021), [arXiv:2103.08591](#).
 - [42] E. Knill, Quantum computing with realistically noisy devices, *Nature* **434**, 39 (2005), [arXiv:quant-ph/0410199](#).
 - [43] E. Fontana, N. Fitzpatrick, D. M. Ramo, R. Duncan, and I. Rungger, Evaluating the noise resilience of variational quantum algorithms, *Physical Review A* **104**, 022403 (2021).
 - [44] M. J. D. Powell, A direct search optimization method that models the objective and constraint functions by linear interpolation, in *Advances in Optimization and Numerical Analysis*, edited by S. Gomez and J.-P. Hennart (Springer, Dordrecht, 1994) pp. 51–67.
 - [45] R. H. Byrd, P. Lu, J. Nocedal, and C. Zhu, A limited memory algorithm for bound constrained optimization, *SIAM Journal on Scientific Computing* **16**, 1190 (1995).
 - [46] E. Fontana, M. S. Rudolph, R. Duncan, I. Rungger, and C. Cirstoiu, Classical simulations of noisy variational quantum circuits, *npj Quantum Information* **11**, 84 (2025), [arXiv:2306.05400](#).
 - [47] S. Thanasilp, S. Wang, M. Cerezo, and Z. Holmes, Exponential concentration in quantum kernel methods, *Nature Communications* **15**, 5200 (2024), [arXiv:2208.11060](#).
 - [48] R. Martínez-Peña and J.-P. Ortega, Quantum reservoir computing in finite dimensions, *Physical Review E* **107**, 035306 (2023), [arXiv:2212.00396](#).
 - [49] R. Takagi, H. Tajima, and M. Gu, Universal sampling lower bounds for quantum error mitigation, *Physical Review Letters* **131**, 210602 (2023).
 - [50] L. Domingo, G. Carlo, and F. Borondo, Taking advantage of noise in quantum reservoir computing, *Scientific Reports* **13**, 8790 (2023), [arXiv:2301.06814](#).
 - [51] F. Monzani, E. Ricci, L. Nigro, and E. Prati, Non-unital noise in a superconducting quantum computer as a computational resource for reservoir computing (2024), [arXiv:2409.07886](#).
 - [52] Z. Zhang, Z. Jia, X. Li, and Y. Dong, A theory of finite-noise optima and generalization in quantum machine learning (2026), [arXiv:2608.24229](#).
 - [53] L. Zhu, Z. Zhang, Y. Dong, and X. Li, Rethinking quantum noise in quantum machine learning: When noise improves learning, *APL Computational Physics* **2**, 030901 (2026), [arXiv:2601.13275](#).
 - [54] F. Scala, G. Guarnieri, and A. Lucchi, Noise-induced equalization in quantum learning models (2025), [arXiv:2511.09428](#).
 - [55] V. Heyraud, Z. Li, Z. Denis, A. Le Boité, and C. Ciuti, Noisy quantum kernel machines, *Physical Review A* **106**, 052421 (2022), [arXiv:2204.12192](#).
 - [56] Y. Du, M.-H. Hsieh, T. Liu, D. Tao, and N. Liu, Quantum noise protects quantum classifiers against adversaries, *Physical Review Research* **3**, 023153 (2021), [arXiv:2003.09416](#).
 - [57] C. Tüysüz, S. Y. Chang, M. Demidik, K. Jansen, S. Vallecorsa, and M. Grossi, Symmetry breaking in geometric quantum machine learning in the presence of noise, *PRX Quantum* **5**, 030314 (2024).
 - [58] H. Ugail and N. Howard, A channel-level diagnostic for symmetry breaking in noisy equivariant quantum neural networks, *IEEE Access* **14**, 95152 (2026).

- [59] J. J. Meyer, M. Mularski, E. Gil-Fuster, A. A. Mele, F. Arzani, A. Wilms, and J. Eisert, Exploiting symmetry in variational quantum machine learning, [PRX Quantum](#) **4**, 010328 (2023), [arXiv:2205.06217](#).
- [60] M. Larocca, F. Sauvage, F. M. Sbahi, G. Verdon, P. J. Coles, and M. Cerezo, Group-invariant quantum machine learning, [PRX Quantum](#) **3**, 030341 (2022), [arXiv:2205.02261](#).
- [61] V.-Q.-M. Nguyen, T.-V. Truong, H.-L. Nguyen, and T.-K. Le, [Code and results for: Zero- versus infinite-temperature damping in variational quantum circuits](#), Zenodo, version v1.0.0 (2026).